

Table of Contents

- **Introduction**
- **Methodology**
- **Analysis**
- **Result**
- **Reference**
- **Appendix**

Introduction

In this report, we are analyzing car sales in Quebec data. Sales are recorded from the year 1960 to 1968. After descriptive analysis of the dataset, we will move on to model specification, model fitting, model validation, and residual analysis, and finally, we will forecast using the best model for 10 points ahead in time.

Methodology

1. R programming software used for analysis
2. Descriptive Analysis
 - 2.1 Data Interpretation
 - 2.2 Data Conversion
 - 2.3 Summary Statistics
 - 2.4 Correlation between lags
 - 2.5 Analysing five points of interest
3. Model Specification
 - 3.1 ACF and PACF plot of raw series
 - 3.2 Normality check for raw series
 - 3.3 Unit-root test of stationarity of raw series
 - 3.4 Transformation (Box-Cox)
 - 3.5 Comparing transformed series and raw series
 - 3.6 Normality check for transformed series
 - 3.7 SARIMA model specification
 - 3.7.1 Specifying Seasonal parameters
 - 3.7.2 Finding possible ARMA parameter values
 - 3.7.2.1 ACF and PACF plot
 - 3.7.2.2 EACF plot
 - 3.7.2.3 BIC table
4. Model Fitting
 - 4.1 Coefficient estimation using CSS, ML, and CSS-ML method
 - 4.2 AIC and BIC score
 - 4.3 Error Measures
5. Residual Analysis
6. Forecasting

Analysis

DESCRIPTIVE ANALYSIS

Reading the CSV file into R

```
# Loading the dataset
setwd("C:/Users/shamb/OneDrive/Desktop/Time Series Analysis/Assignment 3")
data <- read.csv("monthlycarsales.csv")
```

Figure 1

Summary statistics and plot of the dataset

```
head(data)
summary(data)
class(data)
plot(data, type='o', ylab='sales', xlab='Year', main="Cars Sales in Quebec (Monthly)", col="blue")
```

Figure 2

Output

```
> head(data)
      Month Monthly.car.sales.in.Quebec.1960.1968
1 1960.01                                6550
2 1960.02                                8728
3 1960.03                               12026
4 1960.04                               14395
5 1960.05                               14587
6 1960.06                               13791
> summary(data)
      Month Monthly.car.sales.in.Quebec.1960.1968
Min.   :1960 Min.   : 5568
1st Qu.:1962 1st Qu.:11391
Median :1964 Median :14076
Mean   :1964 Mean   :14595
3rd Qu.:1966 3rd Qu.:17596
Max.   :1968 Max.   :26099
> class(data)
[1] "data.frame"
```

Figure 3: Summary statistics of raw series.

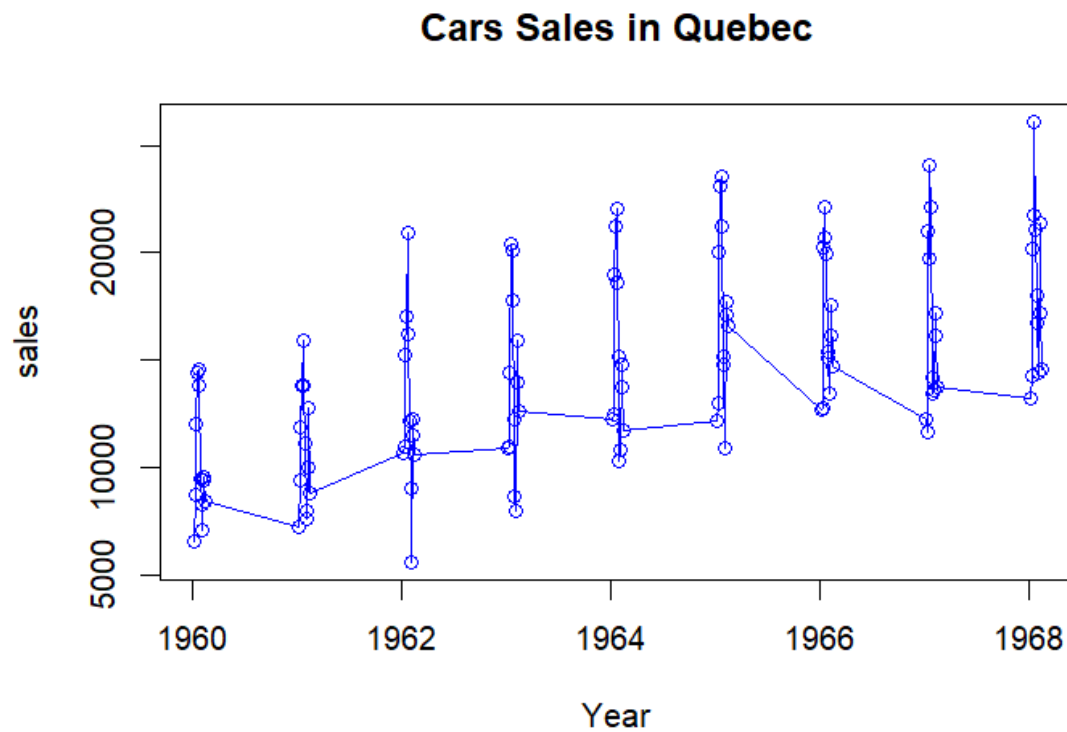


Figure 4: Plot of car sales in Quebec from year 1960 to 1968.

Inference: In Figure 3, the mean monthly car sales value is 14,595. This represents the average sales across all months in the data (from 1960 to 1968) and the median monthly car sales value is 14,076. This indicates that half the month had sales figures higher than 14,076 and the other half had lower sales figures. The minimum sales value (5,568) occurred in 1960, and the maximum (26,099) occurred in 1968. This further strengthens the upward trend, as the highest sales figure within the data range is observed in the most recent year (1968).

Figure 4 shows that the data spans from 1960 to 1968 and is the price of car sales in the Quebec region and supports the summary statistics about an upward trend as there is an increase in car sales.

Converting the data into a time series data

```
# Converting data in a time series object
Car_SalesTS <- ts(data$Monthly.car.sales.in.Quebec.1960.1968, start = c(1960,1),
                  end = c(1968,12), frequency = 12)
```

Figure 5

Analyzing the raw time series data

```
# Descriptive Analysis
class(Car_SalesTS)
summary(Car_SalesTS)
plot(Car_SalesTS, type = 'o', ylab = 'Car Sales', xlab = 'Year',
     main = "Car Sales in Quebec",
     col = "maroon", pch = 15)
```

Figure 6

Output

```
> class(Car_SalesTS)
[1] "ts"
> summary(Car_SalesTS)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  5568   11391   14076   14595   17596   26099
> plot(Car_SalesTS, type = 'o', ylab = 'Car Sales', xlab = 'Year',
+      main = "Car Sales in Quebec",
+      col = "maroon", pch = 15)
```

Figure 7

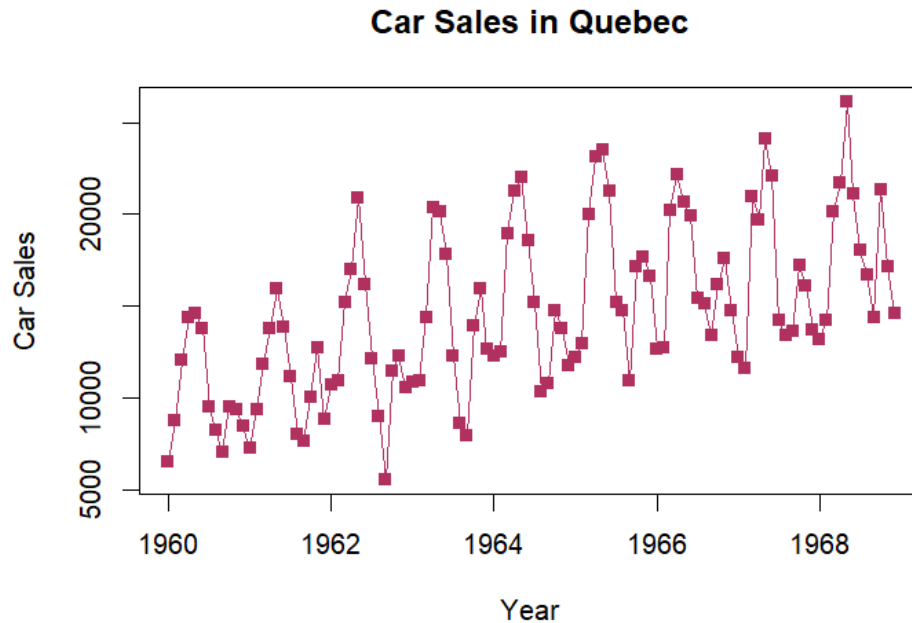


Figure 8: Time Series plot of car sales in Quebec

Inference: In Figure 7 we see the minimum car sales was 5568, Representing the lowest monthly car sales figure observed in the entire dataset. The First Quartile is 11391 - This indicates that 25% of the monthly sales figures fall below 11391 and 75% fall above it. The median car sales is 14076, this is when all monthly sales are arranged in ascending order. Half of the months have sales higher than 14076, and the other half have lower sales. A mean of 14595 represents the average monthly car sales across all months in the data set. The Third Quartile 17596 - This indicates that 75% of the monthly sales figures fall below 17596 and 25% fall above it and the Maximum of 26099 represents the highest monthly car sales figure observed in the data set. The difference between the minimum and maximum values is 20,531 which suggests significant variability. There were months with much lower sales compared to others.

Since the summary statistics are provided for monthly car sales. The variability could be due to seasonal trends (e.g., higher sales during holidays) or other factors. So we plot the time series to confirm our assumption.

Analyzing five points of interest

Figure 8 is the time series plot of our monthly car sales.

1. **Trend:** we see a slightly linear upward trend in the data, indicating an increase in sales or prices of cars from 196- to 1968.
2. **Seasonality:** There are very evident repeating patterns in the data, therefore indicating seasonality
3. **Changepoint or intervention:** There is no apparent change point in the series.

4. **Changing Variance:** By looking at Figure 8 we can assume that at some time points, there is some changing variance, but it is not quite evident.
5. **Behavior:** Since seasonality is quite evident in our time series plot, we cannot say about the behavior of the series

Correlation between consecutive data points

```
# Creating first lag
orig = Car_SalesTS
l = zlag(Car_SalesTS)
index = 2:length(orig)
cor(orig[index],l[index])

# Creating second lag
l2 = zlag(zlag(Car_SalesTS))
index = 3:length(l2)
cor(orig[index],l2[index])
```

Figure 9

Output

```
> cor(orig[index],l[index])
[1] 0.7280681
```

Figure 10: Correlation value between the first lag and original data.

```
> cor(orig[index],l2[index])
[1] 0.3676448
```

Figure 11: Correlation value between the second lag and original data.

Inference: Figures 10 and 11, show the first lag's strong correlation coefficient (0.7281) suggesting that the monthly car sales data exhibits a significant autocorrelation, with each month's sales being heavily impacted by the month before. The second lag's reducing correlation coefficient (0.3676) indicates that although sales patterns are still somewhat persistent, their impact is less significant than it was in the month immediately before. These results suggest that the Car Sales Time Series data exhibits autocorrelation, which is especially high for the initial lag.

Visual representation of the correlation

```
# ACF plot of lags
acf(Car_SalesTS, lag.max = 80)

# Create a scatter plot
plot(x = 1[-1], y = Car_SalesTS[-1], xlab = "Lag", ylab = "Car_SalesTS", col = "green",
     main = "Scatter Plot of Correlations between Lags")
# Add second lag to the plot
points(l2[-c(1, 2)], Car_SalesTS[-c(1, 2)], col = "blue")
legend("topleft", legend = c("First Lag (l)", "Second Lag (l2)"), col = c("green", "blue"), pch = 1)
```

Figure 12

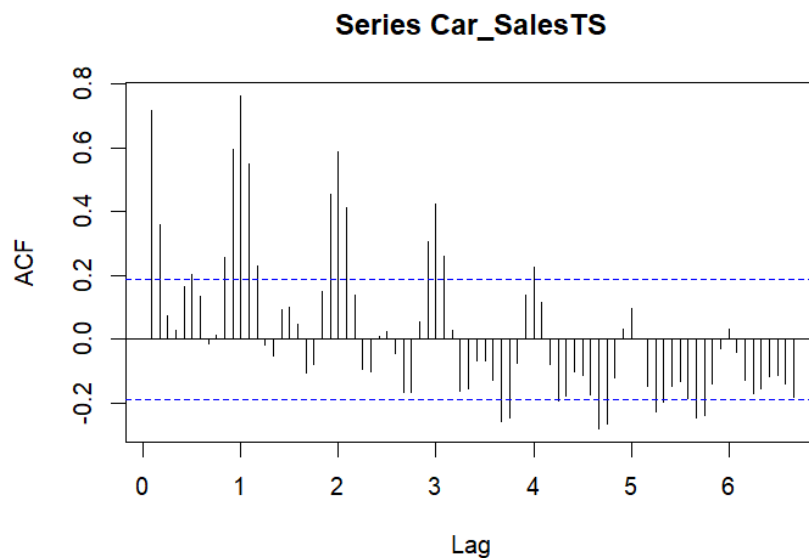


Figure 13: ACF plot of lags.

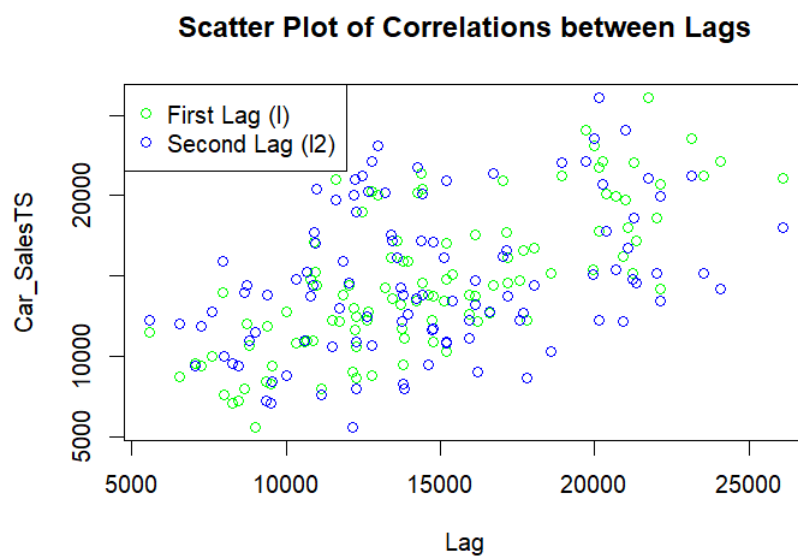


Figure 14: Scatter plot of correlation of first and second lag.

Inference: Figure 13 has significant lags at the seasons and there is a slowly decaying pattern as well, this indicates that the series has seasonality. Hence, our approach for model specification will be SARIMA models. Figure 14 shows the scatter plot, there is no apparent cluster formation, and the data points are scattered all over the plot. Therefore, there isn't a strong correlation between the first and second lag.

MODEL SPECIFICATION

ACF and PACF plot of time series data

```
par(mfrow=c(1,2))
acf(Car_SalesTS, lag.max=80, main ="ACF plot of Car Sales series.")
pacf(Car_SalesTS, lag.max=80, main ="PACF plot of Car Sales series.")
par(mfrow=c(1,1))
```

Figure 15

Output

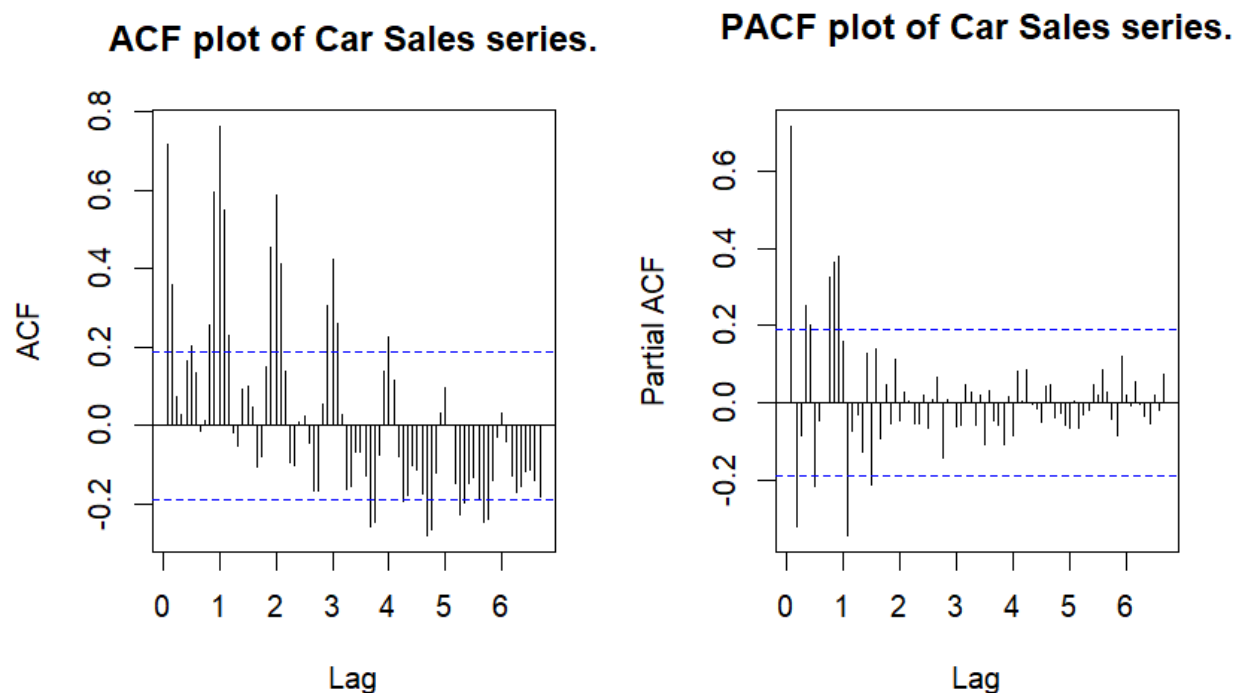


Figure 16: ACF and PACF plot of time series data.

Inference: In Figure 16, we see the ACF and PACF plots of the time series data. The significant lag at 1 and decaying pattern that repeats in the ACF plot strongly suggests a seasonal pattern in the car sales data as the pattern decays over lags indicating that the influence of past seasons weakens as we move further back in time. The repeating aspect of the decaying pattern further reinforces the seasonality. The PACF plot has one highly significant lag which indicates a trend in the series. From both the ACF and PACF plots, it is suggested that the time series has seasonality.

Checking for normality

```
# QQ line
qqnorm(Car_SalesTS, ylab='Total Passengers', xlab='Normal Score'
, main= "QQ Plot of car sales Time Series", col="cornsilk4")
qqline(Car_SalesTS, col = "darkgreen")

# Shapiro Wilk Test
shapiro.test(Car_SalesTS)
```

Figure 17

Output

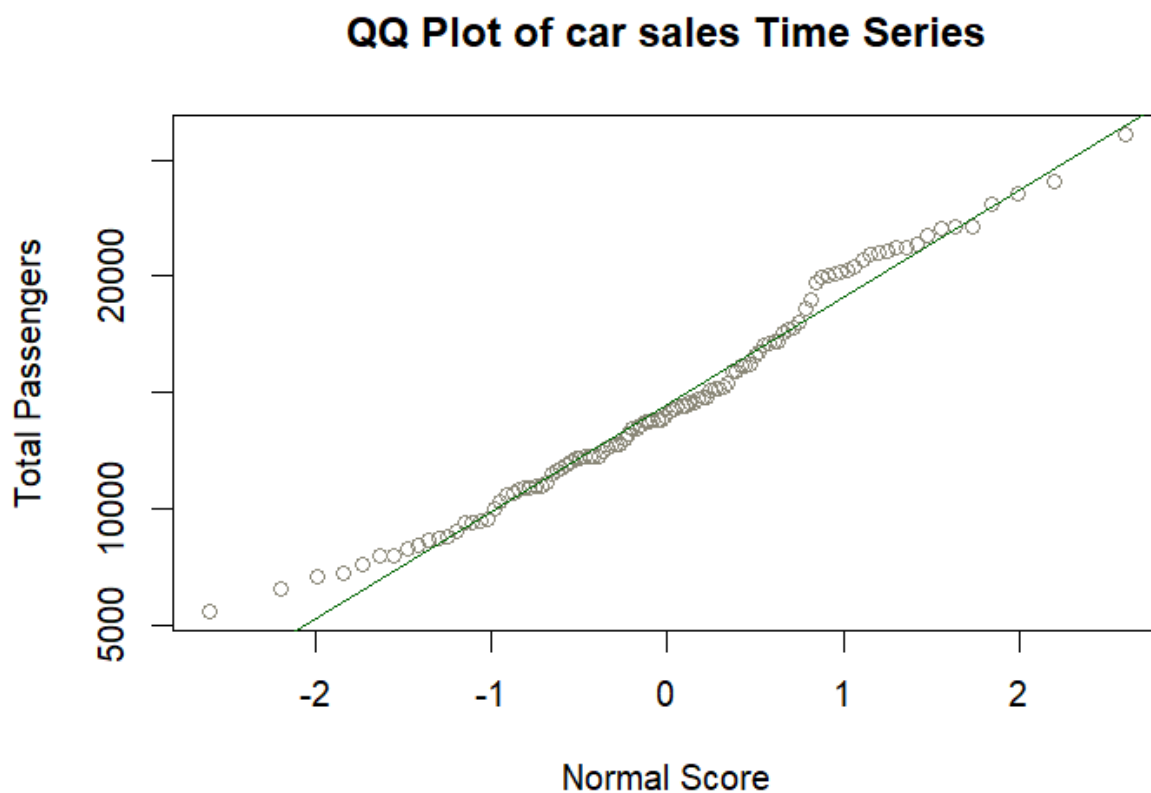


Figure 18: Q-Q plot of the time series.

```
Shapiro-Wilk normality test

data:  Car_SalesTS
W = 0.97746, p-value = 0.06336
```

Figure 19: Shapiro-Wilk test results.

Inference: We see from Figure 18 that the data points are all not captured by the line especially at the ends indicating that the data points in the beginning and ends are not captured and therefore not normal, so we confirm our assumption with the Shapiro-Wilk test, wherein the **Null Hypothesis** (H0) is that the data comes from a normal distribution. Where our threshold value is 0.05 and in Figure 19 we see that the p-value is 0.063 which is greater than 0.05, so when the p-value is greater than the chosen significance level (0.05), we fail to reject the null hypothesis that the data is normally distributed therefore confirming our assumption for figure 18.

Unit root stationarity test

```
# Unitroot Test of the car sales Series
stationarity_tests(Car_SalesTS)
```

Figure 20

Output

```
> stationarity_tests(Car_SalesTS)
Augmented Dickey-Fuller (ADF) Test:

      Augmented Dickey-Fuller Test

data: data
Dickey-Fuller = -4.3241, Lag order = 4, p-value = 0.01
alternative hypothesis: stationary

Kwiatkowski-Phillips-Schmidt-Shin (KPSS) Test:

      KPSS Test for Level Stationarity

data: data
KPSS Level = 1.3313, Truncation lag parameter = 4, p-value = 0.01

Phillips-Perron (PP) Test:

      Phillips-Perron Unit Root Test

data: data
Dickey-Fuller Z(alpha) = -38.583, Truncation lag parameter = 4, p-value = 0.01
alternative hypothesis: stationary

Warning messages:
1: In adf.test(data) : p-value smaller than printed p-value
2: In kpss.test(data) : p-value smaller than printed p-value
3: In pp.test(data) : p-value smaller than printed p-value
```

Figure 21: ADF, PP, and KPSS test results.

Inference: we create a function for checking stationarity called the stationarity_tests() for our car sales data, we have 3 unit root tests that check stationarity in function 1. ADF test, or Augmented Dickey-Fuller test, is a popular unit root test used to assess the stationarity of a time

series where the null hypothesis (H0) of the ADF test states that the time series variable has a unit root, This implies that the variable is non-stationary and exhibits trends or seasonality.

2. The KPSS test is another common unit root test used in time series analysis. It complements the ADF test by taking a different approach to assessing stationarity. Unlike the ADF test, the null hypothesis (H0) of the KPSS test states that the time series variable is stationary and the alternative hypothesis (H1) of the KPSS test suggests the presence of a unit root. implying the variable is non-stationary. 3. The PP test is similar to the ADF test the null hypothesis (H0) of the PP test states that the time series variable has a unit root. This implies the variable is non-stationary.

For all three tests, the Significance level being 0.05 we see that in Figure 21 the values are very small indicating stationarity and rejection of the null hypothesis for the ADF and PP test and the KPSS test indicates the presence of unit root, which is not considered since 2 out of three tests indicated stationarity.

Transformation

```
# Box-Cox Transformation on the time series to minimize changing variance
BC = BoxCox.ar(Car_SalesTS)
title(main= "Log-likelihood Vs Values of Lambda for car sales Series")
BC$ci
lambda ← BC$lambda[which(max(BC$loglike) == BC$loglike)]
lambda
Car_SalesTS_BC = (Car_SalesTS^lambda-1)/lambda
```

Figure 22

Output

```
> BC$ci
[1] 0.3 1.2
> lambda ← BC$lambda[which(max(BC$loglike) == BC$loglike)]
> lambda
[1] 0.8
> Car_SalesTS_BC = (Car_SalesTS^lambda-1)/lambda
```

Figure 23: Box-Cox interval and lambda value.

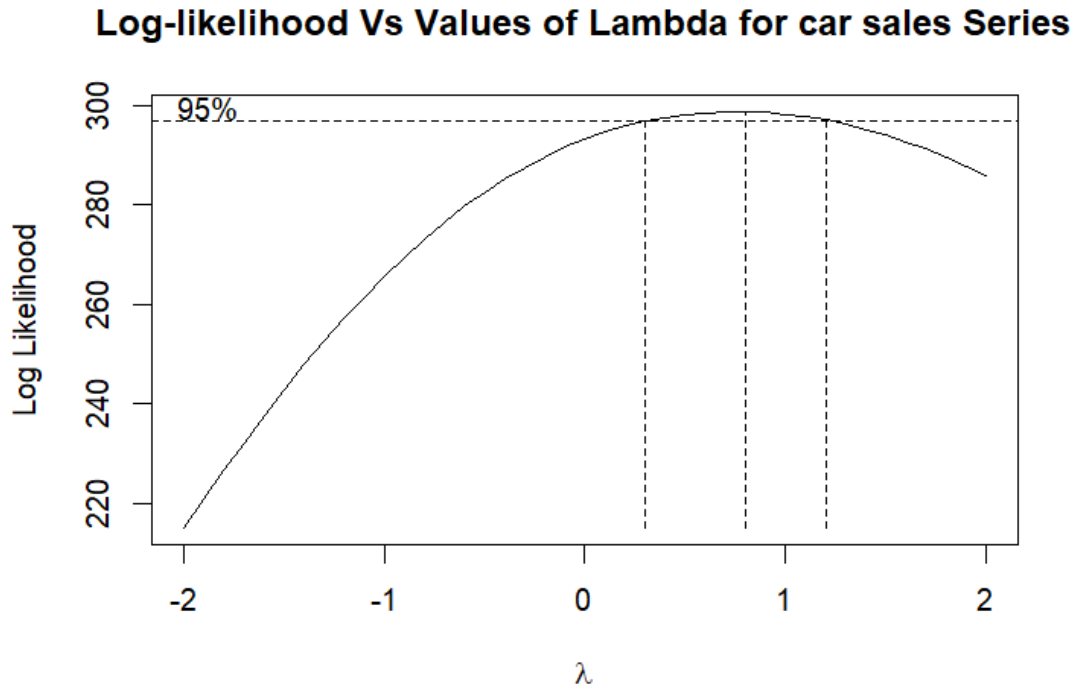


Figure 24: Box-Cox plot

Inference: Figures 23 and 24, the confidence interval $[0.3, 1.2]$ suggests that the estimated parameter lambda is reasonably certain to be within this range. The lambda value of 0.8, selected based on maximizing the log-likelihood, indicates that the Box-Cox transformation applied to the data might be with $\lambda = 0.8$. The transformation indicated by the chosen lambda value of 0.8 lies between a simple transformation ($\lambda = 1$) and a square root ($\lambda = 0.5$). To be more precise, $\lambda = 0.8$ indicates a less powerful transformation that is more similar to a square root transformation.

Visualizing Transformed Series Vs Original Global Temperature Series

```
par(mfrow=c(2,1))
plot(Car_SalesTS_BC, type = 'o', ylab = 'Car Sales', xlab = 'Year',
     main = "Transformed Time series plot of Car Sales",
     col = "Blue", pch = 10)
plot(Car_SalesTS, type = 'o', ylab = 'Car Sales', xlab = 'Year',
     main = "Time series plot of Car Sales",
     col = "maroon", pch = 10)
```

Figure 25

Output

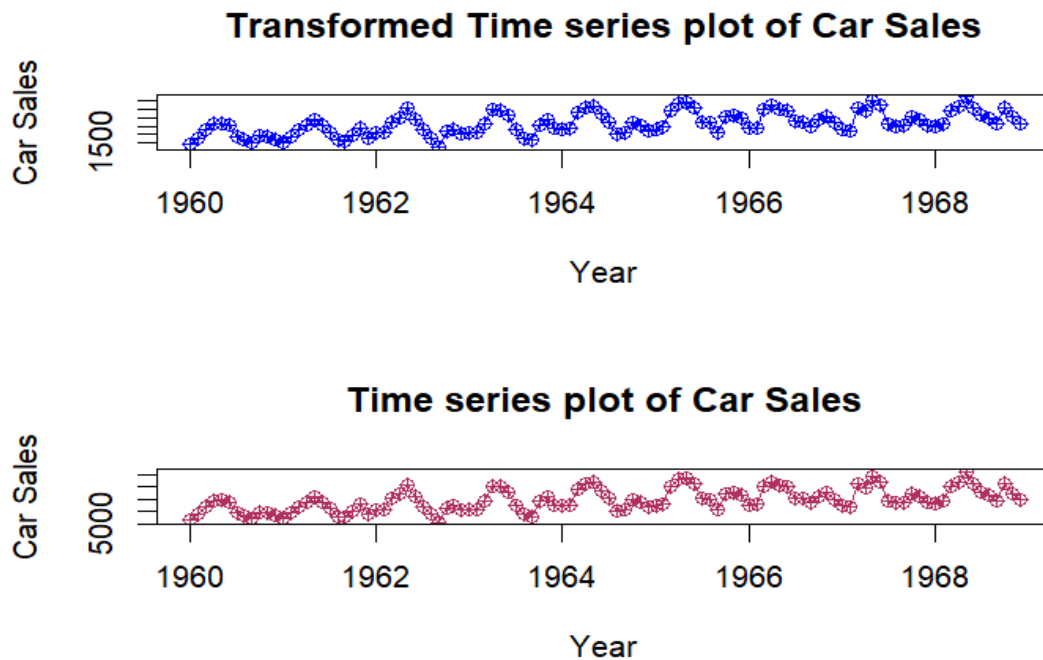


Figure 26: Comparison of transformed series and the raw time series plot.

Inference: From Figure 26 the time series plot of the Box-Cox transferred series and the raw time series look approximately the same. But we cannot say for sure so we will confirm if the transformation is improving the normality of the series by undertaking the Shapiro-Wilk normality test and by looking at the Q-Q plots.

Checking for normality in the transformed time series

```
# QQ line
par(mfrow=c(2,1))
qqnorm(Car_SalesTS_BC, ylab="Box-Cox GlobalTemp_ts", xlab="Normal Scores",
       col= 'darkcyan',
       main="QQ Plot of Transformed Global Temperature Time Series")
qqline(Car_SalesTS_BC, col = 'darkred')
qqnorm(Car_SalesTS, ylab='Total Passengers', xlab='Normal Score',
       main= "QQ Plot of US Air Passengers Time Series", col="cornsilk4")
qqline(Car_SalesTS, col = "darkgreen")

# Shapiro Wilk Test
shapiro.test(Car_SalesTS_BC)
```

Figure 27

Output

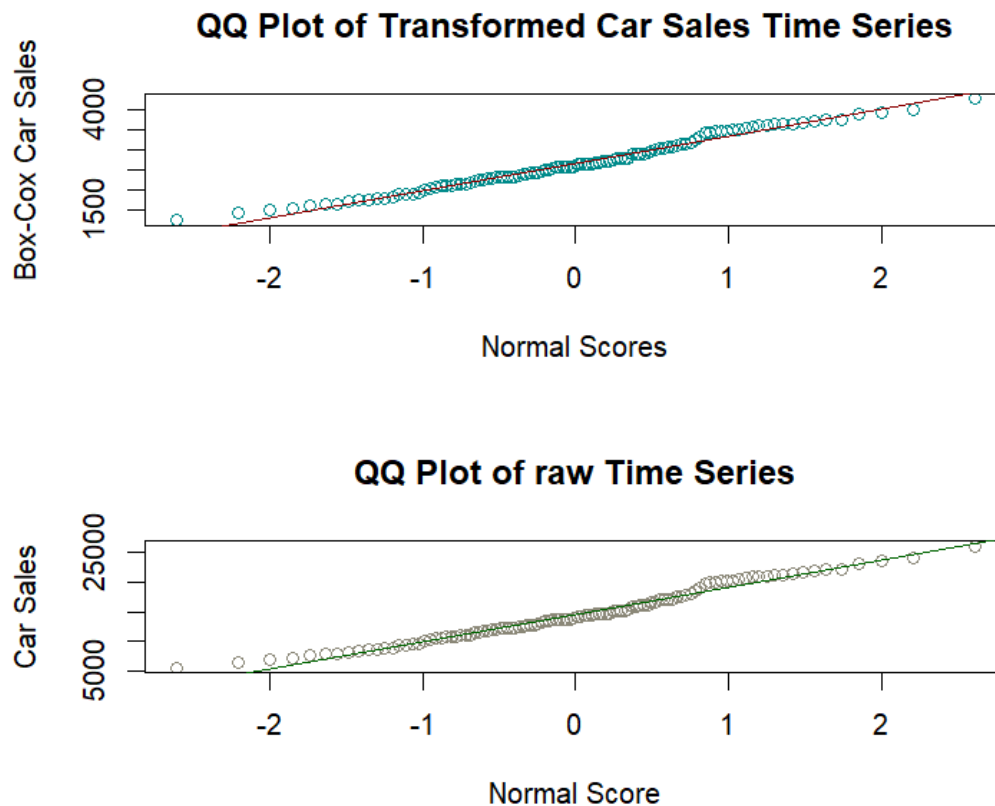


Figure 28: Q-Q plot of transformed and raw car sales series.

<pre>> shapiro.test(Car_SalesTS_BC)</pre>	Shapiro-Wilk normality test
Shapiro-Wilk normality test	
data: Car_SalesTS_BC	data: Car_SalesTS
W = 0.98306, p-value = 0.1864	W = 0.97746, p-value = 0.06336

Figure 29: Comparison of Shapiro-Wilk test results of transformed series and raw time series.

Inference: Figure 28 shows how similar the two Q-Q plots appear, making it impossible to determine from the plots whether or not the transformation improved the normality. On the other hand, Figure 29 shows that the raw time series' p-value (0.06336) is just above the usual significance limit of 0.05. This implies that there isn't much evidence to reject the null hypothesis, but the transformed series produced a p-value of 0.1864, which is noticeably higher than 0.05. This suggests that the data distribution's normality has been enhanced via the Box-Cox treatment. Therefore, based on the Shapiro-Wilk tests and considerations of normality, the Box-Cox transformed data is recommended for further analysis.

SARIMA model specification

```
m1.car_sales = Arima(Car_SalesTS_BC,order=c(0,0,0),seasonal=list(order=c(0,1,0), period=12))
res.m1 = residuals(m1.car_sales);
par(mfrow=c(1,1))
plot(res.m1,xlab='Time',ylab='Residuals',main="Time series plot of the residuals")
par(mfrow=c(1,2))
acf(res.m1, lag.max = 80, main = "The sample ACF of the residuals")
pacf(res.m1, lag.max = 80, main = "The sample PACF of the residuals")
```

Figure 30

Output

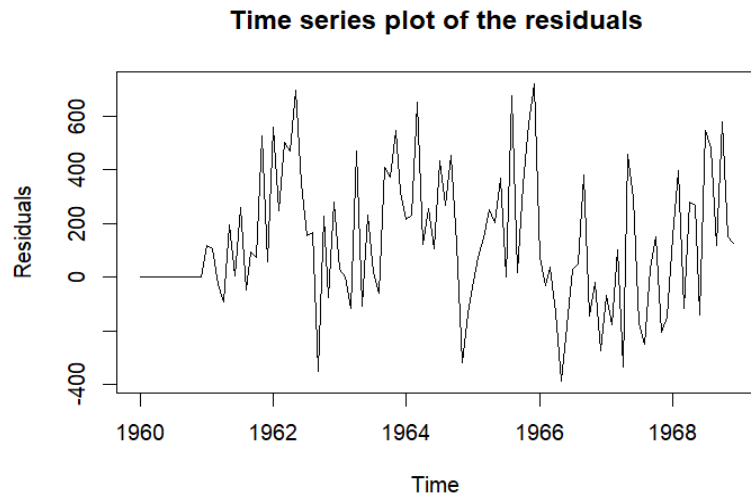


Figure 31: Time series plot of residuals of m1 model.

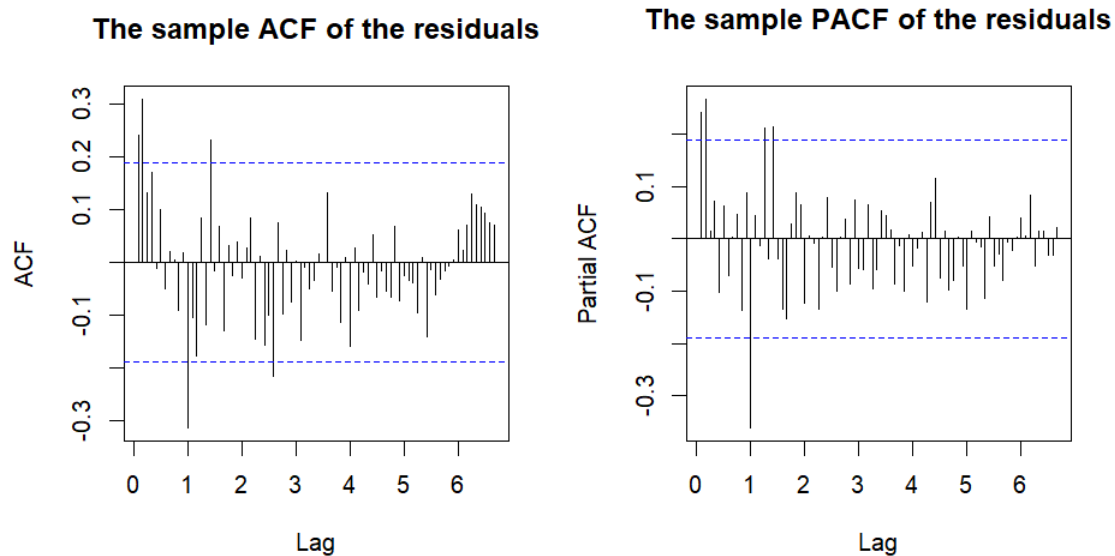


Figure 32: ACF and PACF plot of m1 model.

Inference: When doing model specification for the SARIMA model we initially take $D=1$, to get rid of the seasonality in the series. P and Q are kept as 0 as of now. Figure 32 shows that there is a significant seasonal lag at season 1 in the ACF and PACF plots. To get rid of these seasonal lags we will increase the values of parameters P and Q .

```
m2.car_sales = Arima(Car_SalesTS_BC,order=c(0,0,0),seasonal=list(order=c(1,1,1), period=12))
res.m2 = residuals(m2.car_sales);
par(mfrow=c(1,1))
plot(res.m2,xlab='Time',ylab='Residuals',main="Time series plot of the residuals")
par(mfrow=c(1,2))
acf(res.m2, lag.max = 80, main = "The sample ACF of the residuals")
pacf(res.m2, lag.max = 80, main = "The sample PACF of the residuals")
```

Figure 33

Output

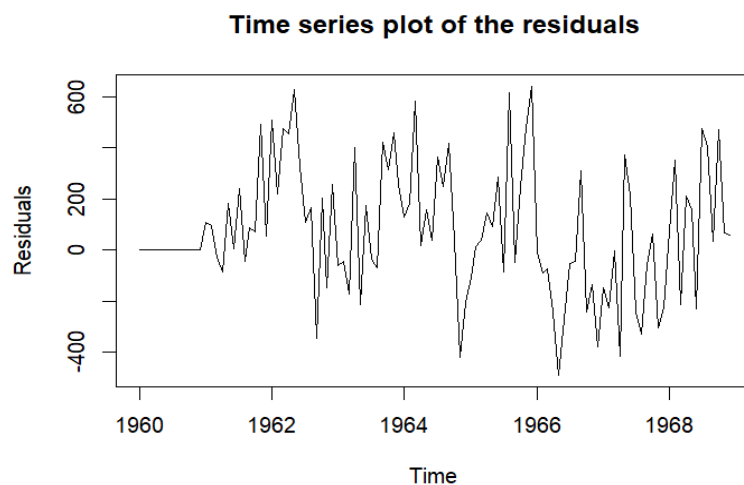


Figure 34: Time series plot of residuals of m2 model.

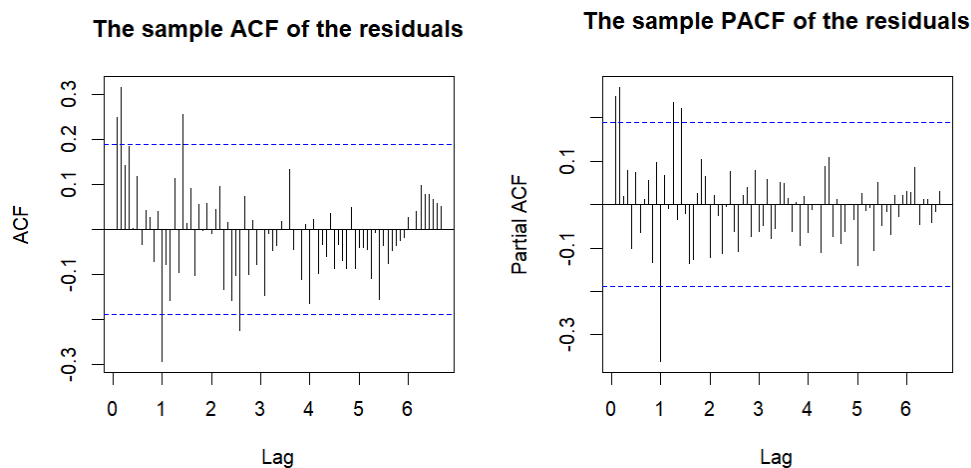


Figure 35: ACF and PACF plot of m2 model.

Inference: After setting $P=1$ and $Q=1$ we still have those significant lags although as seen in Figure 35 the lag values have slightly decreased. We will increase the values of seasonal parameters to completely get rid of the seasonal lags.

```
m3.car_sales = Arima(Car_SalesTS_BC,order=c(0,0,0),seasonal=list(order=c(2,1,1), period=12))
res.m3 = residuals(m3.car_sales);
par(mfrow=c(1,1))
plot(res.m3,xlab='Time',ylab='Residuals',main="Time series plot of the residuals")
par(mfrow=c(1,2))
acf(res.m3, lag.max = 80, main = "The sample ACF of the residuals")
pacf(res.m3, lag.max = 80, main = "The sample PACF of the residuals")
```

Figure 36

Output

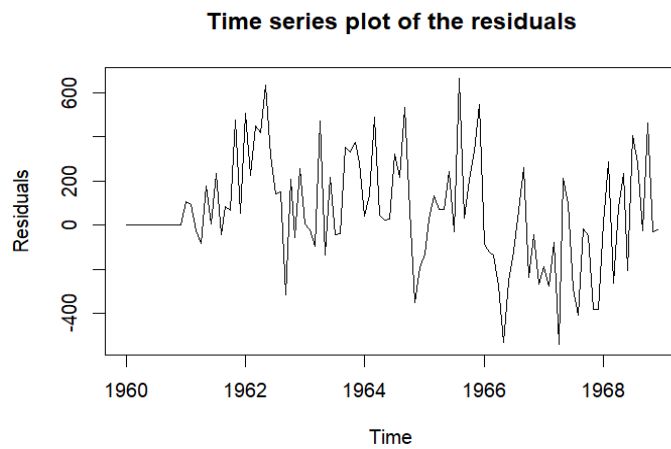


Figure 37: Time series residual plot of m3 plot.

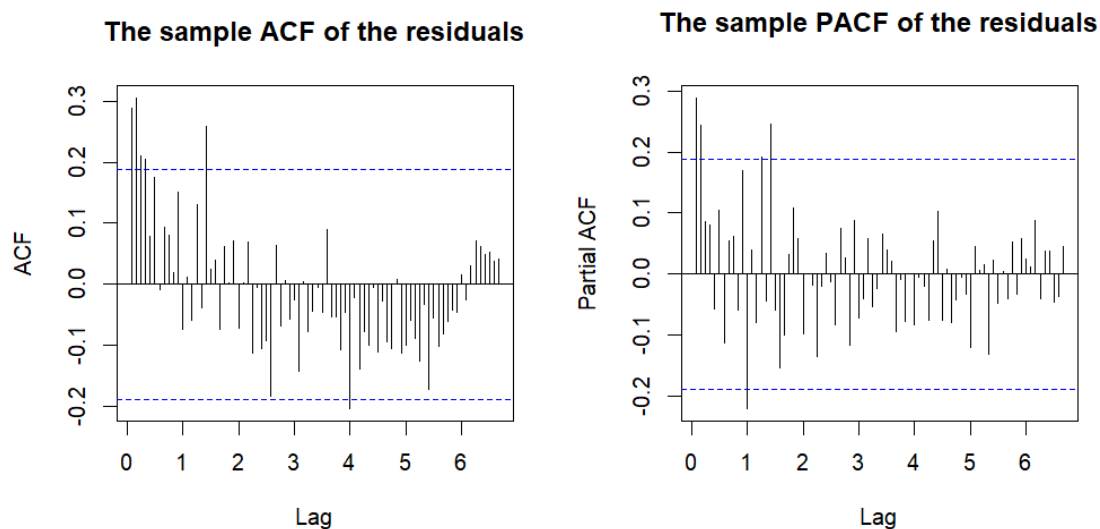


Figure 38: ACF and PACF plot of m3 model.

Inference: When we take seasonal parameters $P=2$, we get rid of the seasonal lag in the ACF plot although the seasonal lag in the PACF plot remains, rather than increasing the seasonal parameters now as that might lead to overfitting of the seasonal part, we will perform a stationarity test on the m3 model and then specifying the ordinary p, d, and q parameters. The time series plot of residual has an almost constant mean level.

Checking for stationarity of the m3 model

```
stationarity_tests(res.m3)
```

Figure 39

Output

```
> stationarity_tests(res.m3)
Augmented Dickey-Fuller (ADF) Test:

      Augmented Dickey-Fuller Test

data: data
Dickey-Fuller = -3.6172, Lag order = 4, p-value = 0.03493
alternative hypothesis: stationary

Kwiatkowski-Phillips-Schmidt-Shin (KPSS) Test:

      KPSS Test for Level Stationarity

data: data
KPSS Level = 0.41468, Truncation lag parameter = 4, p-value = 0.07083
```

Figure 40: ADF and KPSS test of m3 model.

```
Phillips-Perron (PP) Test:

      Phillips-Perron Unit Root Test

data: data
Dickey-Fuller Z(alpha) = -91.629, Truncation lag parameter = 4, p-value = 0.01
alternative hypothesis: stationary
```

Figure 41: pp- test of m3 model.

Inference: After doing the stationarity test on the m3 model, a p-value of 0.03493 is obtained from the Augmented Dickey-Fuller (ADF) test, which is less than the significance level of 0.05. As a result, we determine that the time series is most likely stationary and reject the null hypothesis. With a p-value of 0.07083, larger than 0.05, obtained from the KPSS test, we are unable to reject the null hypothesis and come to the conclusion that the series is stationary. The p-value for the Phillips-Perron (PP) test is 0.01. Since the p-value is less than 0.05, the null hypothesis is rejected, leading us to believe that the time series is most likely stationary. After doing the stationarity test, we find that the value of $d=0$.

Ordinary p and q values will be specified by looking at the ACF and PACF plots of the m3 model. The significant lags before season 1 in the ACF plot are 4 and in the PACF plot are 2. Hence, $p=2$ and $q=4$.

```
m4.car_sales = Arima(Car_SalesTS_BC,order=c(2,0,4),seasonal=list(order=c(2,1,1), period=12))
res.m4 = residuals(m4.car_sales);
par(mfrow=c(1,1))
plot(res.m4,xlab='Time',ylab='Residuals',main="Time series plot of the residuals")
par(mfrow=c(1,2))
acf(res.m4, lag.max = 80, main = "The sample ACF of the residuals")
pacf(res.m4, lag.max = 80, main = "The sample PACF of the residuals")
```

Figure 42

Output

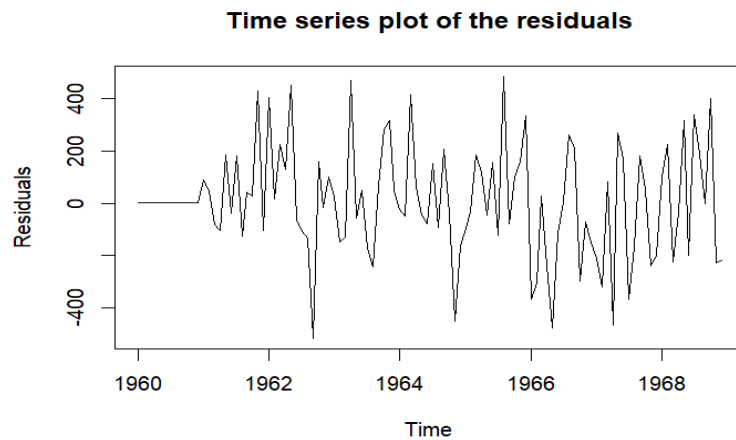


Figure 43: Time series residual plot of m4 model.

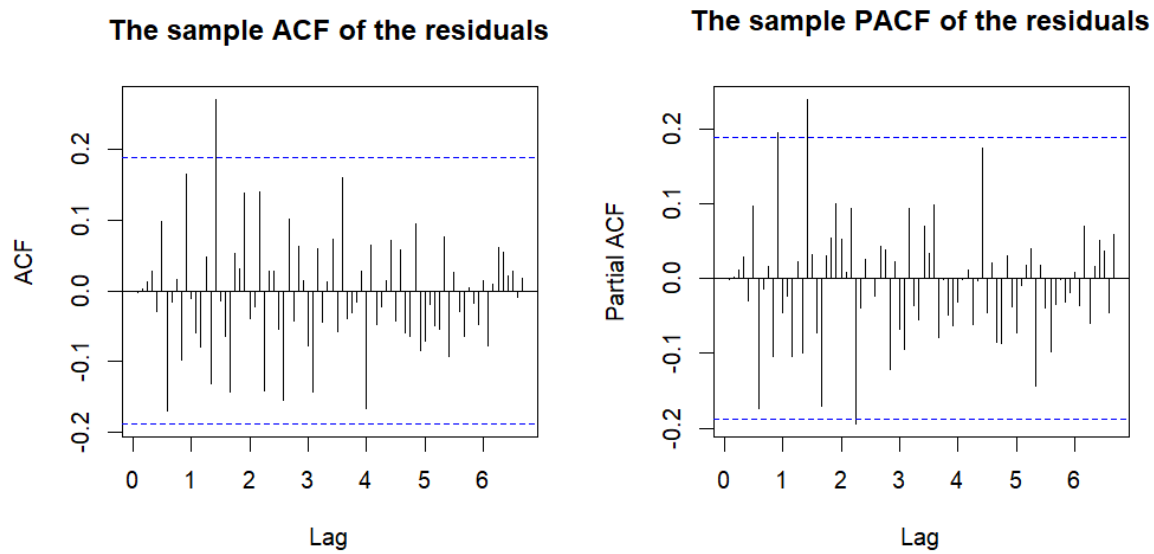


Figure 44: ACF and PACF plots of the m4 model.

Inference: Figure 44 shows that once we specify the ordinary p, d, and q in the model specification. We get rid of all the seasonal lags in both the ACF and PACF plots. The time series plot of residuals also looks stationary with a constant mean level. Hence, the final values of parameters are the seasonal part being P=2, D=1, and Q=1 and the ordinary parts are p=2, d=0, and q=4.

Model specified is SARIMA(2,0,4)x(2,1,1)₁₂

Specifying parameters p,d, and q from the EACF plot

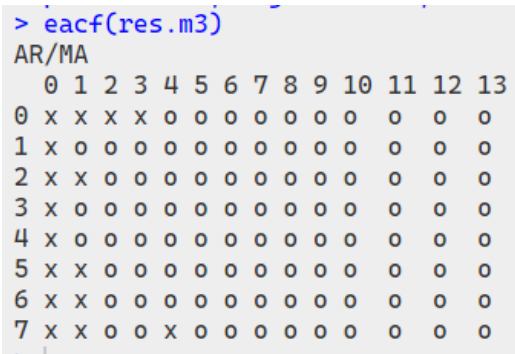


Figure 45: EACF plot.

Inference: In Figure 45, the first 0 not interrupted by x is at (1,1).

The set of possible SARIMA models from EACF are:

SARIMA(1,0,1)x(2,1,1)₁₂

SARIMA(1,0,2)x(2,1,1)₁₂

SARIMA(2,0,2)x(2,1,1)₁₂

Specifying parameters p,d, and from the BIC table

```
par(mfrow=c(1,1))
bic_table = armasubsets(y=res.m3,nar=5,nma=5,y.name='p',ar.method='ols')
plot(bic_table)
```

Figure 46

Output

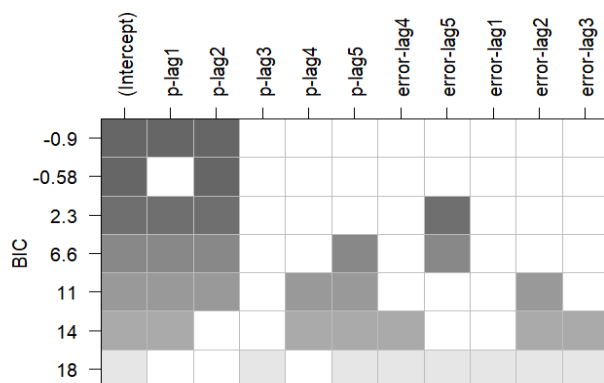


Figure 47: BIC table.

Inference: From Figure 47 which is the BIC table we will take values of p as 1 and 2 as these values are supported by the best models. Whereas for values of q , there are no parameters that are supported by the top two best models hence $q = 0$.

The set of possible SARIMA models from the BIC table are:

SARIMA(1,0,0)x(2,1,1)₁₂

SARIMA(2,0,0)x(2,1,1)₁₂

All possible SARIMA models from ACF, PACF, EACF, and BIC table are as follows:

SARIMA(2,0,4)x(2,1,1)₁₂

SARIMA(1,0,1)x(2,1,1)₁₂

SARIMA(1,0,2)x(2,1,1)₁₂

SARIMA(2,0,2)x(2,1,1)₁₂

SARIMA(1,0,0)x(2,1,1)₁₂

SARIMA(2,0,0)x(2,1,1)₁₂

MODEL FITTING

Coefficient estimation using ML, CSS, and CSS-ML method

```
# Parameter Estimation
orders <- list(c(2,0,4), c(1,0,1), c(1,0,2), c(2,0,2), c(1,0,0), c(2,0,0))
seasonal_orders <- list(c(2,1,1))
results <- fit_sarima_models(Car_SalesTS_BC, orders, seasonal_orders, period=12)

# Print results
for (name in names(results)) {
  cat("\nModel:", name, "\n")
  if (!is.null(results[[name]])) {
    print(results[[name]])
  } else {
    cat("No results for this model due to an error.\n")
  }
}
```

Figure 48: Coef test of the specified models.

Inference:

Models	CSS	ML	CSS-ML
SARIMA(2,0,4)x(2,1,1) _12	ar1, sar1 significant.	ar2, ma2, sar1, sma1 significant.	ar1 significant.
SARIMA(1,0,1)x(2,1,1) _12	ar1, ma1, sar1 significant.	ar1, ma1, sar1, sma1 significant.	ar1, ma1, sar1, sma1 significant.
SARIMA(1,0,2)x(2,1,1) _12	ar1, ma1, ma2, sar1, sar2 significant.	ar1, ma1, sar1, sma1 significant.	ar1, ma1, sar1, sma1 significant.
SARIMA(2,0,2)X(2,1,1) _12	ar1 significant.	ar2, ma2, sar1, sma1 significant.	Error in running CSS- ML.
SARIMA(1,0,0)x(2,1,1) _12	ar1, sar1, sar2 significant.	ar1, sar1, sar2, sma1 significant.	ar1, sar1, sar2, sma1 significant.
SARIMA(2,0,0)x(2,1,1) _12	ar1 significant.	ar1, ar2 significant.	ar1, ar2 significant.

Table 1: Coefficient estimation summary.

Table 1 demonstrates the reliability of the SARIMA(1,0,1)x(2,1,1)₁₂ and SARIMA(1,0,2)x(2,1,1)₁₂ models by demonstrating high significance for many parameters across all techniques i.e. CSS, ML and CSS-ML. For time series forecasting, these models with stable parameter significance are therefore probably the best out of all the specified models.

AIC and BIC score of SARIMA models

```
# AIC and BIC score

sc.AIC = AIC(m3_204.salesML, m3_101.salesML, m3_102.salesML,
             m3_202.salesML, m3_100.salesML, m3_200.salesML)

sc.BIC = BIC(m3_204.salesML, m3_101.salesML, m3_102.salesML,
             m3_202.salesML, m3_100.salesML, m3_200.salesML)

sort.score(sc.AIC, score = "aic")
sort.score(sc.BIC, score = "bic")
```

Figure 49

<pre>> sort.score(sc.AIC, score = "aic")</pre> <table border="1"> <thead> <tr> <th></th> <th>df</th> <th>AIC</th> </tr> </thead> <tbody> <tr> <td>m3_101.salesML</td> <td>6</td> <td>1326.980</td> </tr> <tr> <td>m3_102.salesML</td> <td>7</td> <td>1328.269</td> </tr> <tr> <td>m3_202.salesML</td> <td>8</td> <td>1329.615</td> </tr> <tr> <td>m3_204.salesML</td> <td>10</td> <td>1330.061</td> </tr> <tr> <td>m3_200.salesML</td> <td>6</td> <td>1336.549</td> </tr> <tr> <td>m3_100.salesML</td> <td>5</td> <td>1345.774</td> </tr> </tbody> </table>		df	AIC	m3_101.salesML	6	1326.980	m3_102.salesML	7	1328.269	m3_202.salesML	8	1329.615	m3_204.salesML	10	1330.061	m3_200.salesML	6	1336.549	m3_100.salesML	5	1345.774	<pre>> sort.score(sc.BIC, score = "bic")</pre> <table border="1"> <thead> <tr> <th></th> <th>df</th> <th>BIC</th> </tr> </thead> <tbody> <tr> <td>m3_101.salesML</td> <td>6</td> <td>1342.367</td> </tr> <tr> <td>m3_102.salesML</td> <td>7</td> <td>1346.219</td> </tr> <tr> <td>m3_202.salesML</td> <td>8</td> <td>1350.130</td> </tr> <tr> <td>m3_200.salesML</td> <td>6</td> <td>1351.935</td> </tr> <tr> <td>m3_204.salesML</td> <td>10</td> <td>1355.705</td> </tr> <tr> <td>m3_100.salesML</td> <td>5</td> <td>1358.596</td> </tr> </tbody> </table>		df	BIC	m3_101.salesML	6	1342.367	m3_102.salesML	7	1346.219	m3_202.salesML	8	1350.130	m3_200.salesML	6	1351.935	m3_204.salesML	10	1355.705	m3_100.salesML	5	1358.596
	df	AIC																																									
m3_101.salesML	6	1326.980																																									
m3_102.salesML	7	1328.269																																									
m3_202.salesML	8	1329.615																																									
m3_204.salesML	10	1330.061																																									
m3_200.salesML	6	1336.549																																									
m3_100.salesML	5	1345.774																																									
	df	BIC																																									
m3_101.salesML	6	1342.367																																									
m3_102.salesML	7	1346.219																																									
m3_202.salesML	8	1350.130																																									
m3_200.salesML	6	1351.935																																									
m3_204.salesML	10	1355.705																																									
m3_100.salesML	5	1358.596																																									

Figure 50: AIC and BIC scores of all the specified models.

AIC Scores: Figure 50 shows us the AIC score of all possible models. With an AIC of 1326.980, SARIMA (1,0,1)x(2,1,1)₁₂ has the lowest score. SARIMA (1,0,2)x(2,1,1)₁₂ and SARIMA (2,0,2)x(2,1,1)₁₂, with AIC values of 1328.269 and 1329.615, respectively, are the next best models according to AIC scores. So, from the AIC score, we will consider:

SARIMA (1,0,1)x(2,1,1)₁₂

SARIMA (1,0,2)x(2,1,1)₁₂

SARIMA (2,0,2)x(2,1,1)₁₂

BIC Scores: Figure 50 shows us the BIC score of all possible models. SARIMA (1,0,1)x(2,1,1)₁₂ has the lowest score, with a BIC of 1342.367. SARIMA (1,0,2)x(2,1,1)₁₂ and SARIMA (2,0,2)x(2,1,1)₁₂, with BIC values of 1346.219 and 1350.130, respectively, are the next best models according to BIC scores. Hence, we will consider:

SARIMA (1,0,1)x(2,1,1)_12

SARIMA (1,0,2)x(2,1,1)_12

SARIMA (2,0,2)x(2,1,1)_12

AIC and BIC scores are consistent for SARIMA (1,0,1)x(2,1,1)_12, SARIMA (1,0,2)x(2,1,1)_12 and SARIMA (2,0,2)x(2,1,1)_12.

Error measures of SARIMA models

```
# Error Measure
Sm3_204.sales <- accuracy(m3_204.salesCSS)[1:7]
Sm3_101.sales <- accuracy(m3_101.salesCSS)[1:7]
Sm3_102.sales <- accuracy(m3_102.salesCSS)[1:7]
Sm3_202.sales <- accuracy(m3_202.salesCSS)[1:7]
Sm3_100.sales <- accuracy(m3_100.salesCSS)[1:7]
Sm3_200.sales <- accuracy(m3_200.salesCSS)[1:7]

df.Smodels <- data.frame(rbind( Sm3_204.sales, Sm3_101.sales, Sm3_102.sales,
                                Sm3_202.sales, Sm3_100.sales, Sm3_200.sales))

colnames(df.Smodels) <- c("ME", "RMSE", "MAE", "MPE", "MAPE",
                          "MASE", "ACF1")
rownames(df.Smodels) <- c("SARIMA(2,0,4)x(2,1,1)_12", "SARIMA(1,0,1)x(2,1,1)_12",
                          "SARIMA(1,0,2)x(2,1,1)_12", "SARIMA(2,0,2)x(2,1,1)_12",
                          "SARIMA(1,0,0)x(2,1,1)_12", "SARIMA(2,0,0)x(2,1,1)_12")
round(df.Smodels, digits = 3)
```

Figure 51

Output

	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1
SARIMA(2,0,4)x(2,1,1)_12	10.242	184.667	120.112	0.121	4.185	0.512	0.002
SARIMA(1,0,1)x(2,1,1)_12	9.000	185.759	124.918	0.039	4.396	0.532	0.112
SARIMA(1,0,2)x(2,1,1)_12	1.498	169.709	109.457	-0.250	3.887	0.466	-0.043
SARIMA(2,0,2)x(2,1,1)_12	11.492	185.740	121.283	0.146	4.229	0.517	0.011
SARIMA(1,0,0)x(2,1,1)_12	48.120	206.760	133.110	1.521	4.562	0.567	-0.239
SARIMA(2,0,0)x(2,1,1)_12	35.107	197.762	131.866	1.038	4.571	0.562	-0.120

Figure 52: Error measure of SARIMA models

Inference: From Figure 52 we find that with a value of 1.498, SARIMA (1,0,2)x(2,1,1)_12 has the lowest ME, indicating less bias in the predictions. With an RMSE of 169.709, the SARIMA (1,0,2)x(2,1,1)_12 model also has the lowest average squared difference (RMSE) between the observed and predicted values. SARIMA (1,0,2)x(2,1,1)_12 model has the lowest MAE of 109.457, meaning it has the fewest average absolute differences. SARIMA (1,0,2)x(2,1,1)_12 has an MPE of -0.250, indicating a slight negative bias. SARIMA (1,0,2)x(2,1,1)_12 has the lowest MAPE (3.887), suggesting the least relative error. SARIMA (1,0,2)x(2,1,1)_12 has the

lowest MASE of all the models, 0.466. For SARIMA (1,0,2)x(2,1,1)_12, the ACF1 is -0.043, meaning that there is little connection between the residuals at lag 1.

SARIMA (1,0,2)x(2,1,1)_12 scores well across the majority of metrics (ME, RMSE, MAE, MPE, MAPE, MASE, and ACF1), which suggests it is probably the most dependable forecasting model.

RESIDUAL ANALYSIS

After model fitting, we get that models

SARIMA (1,0,1)x(2,1,1)_12

SARIMA (1,0,2)x(2,1,1)_12

SARIMA (2,0,2)x(2,1,1)_12

are most likely the best models out of all the tentatively selected models. Error measure states that SARIMA (1,0,2)x(2,1,1)_12 is the most dependable model out of all. We will do a diagnostic check for these three models to choose our final best-fit model for forecasting.

SARIMA(1,0,1)X(2,1,1)

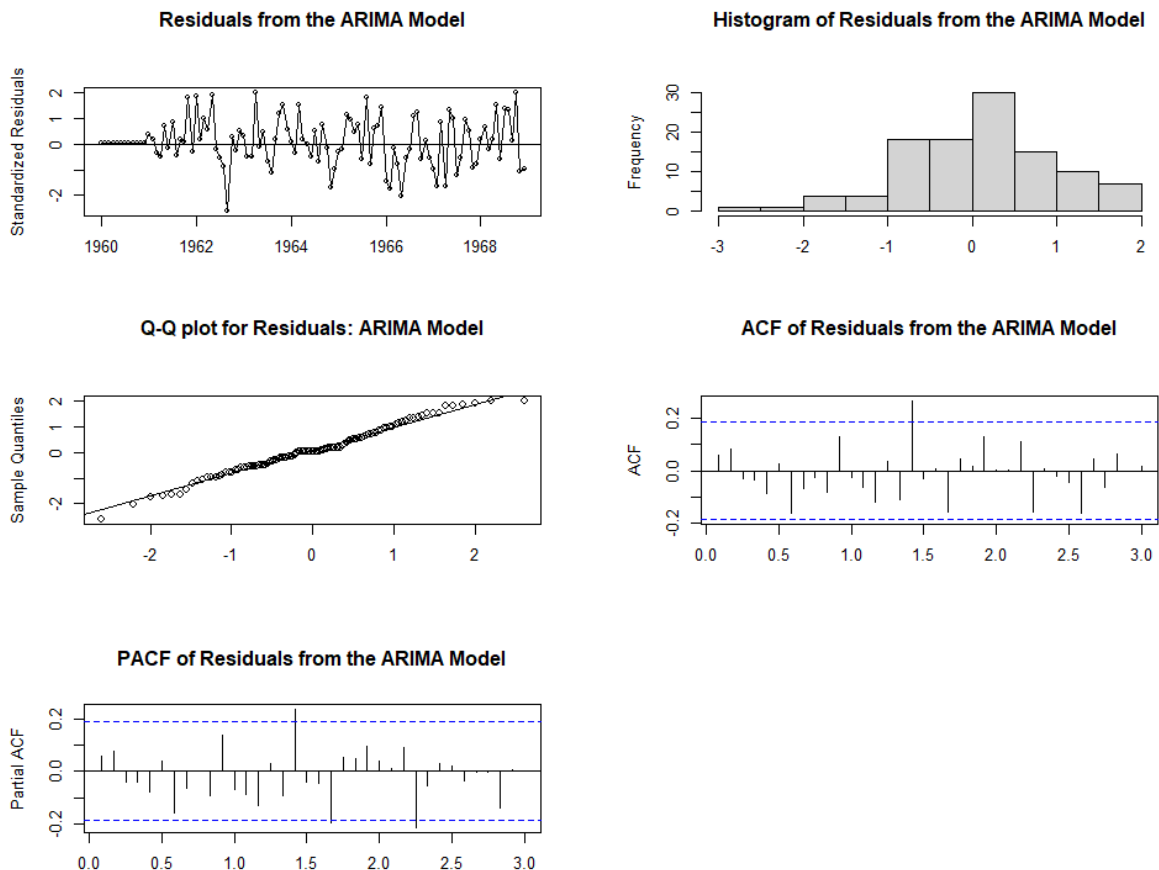


Figure 53: Residual Plots

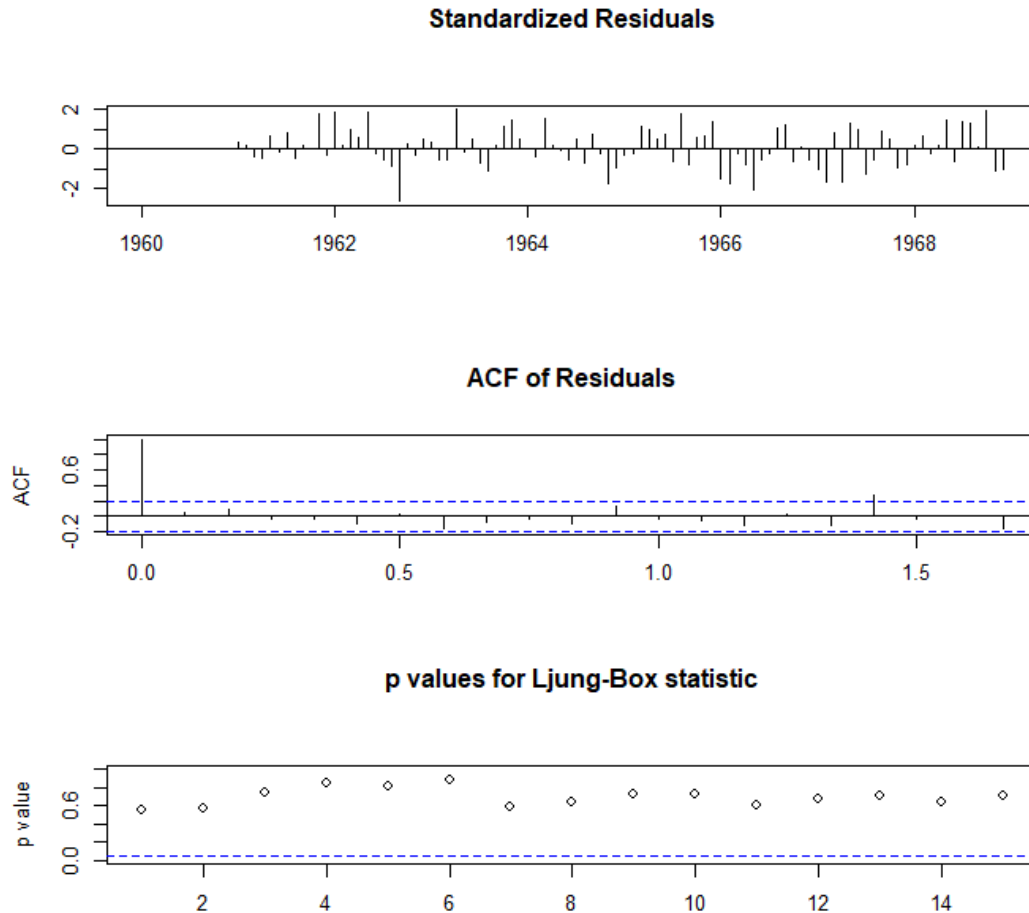


Figure 54: Residual Plots

Shapiro-Wilk normality test	Box-Ljung test
data: rstandard(model) W = 0.98672, p-value = 0.3636	data: rstandard(model) X-squared = 0.3672, df = 1, p-value = 0.5445

Figure 55: Shapiro-Wilk test and Box-Ljung test values.

Inference: Figures 53 and 54 tell us that this model appears to have stationarity, however, the histogram is slightly skewed to the left which suggests that the model lacks symmetry, and the QQ plot and QQ line are aligned. There is one significant late lag in both ACF & PACF. Figure 55 shows the Shapiro-Wilk test, since the p-value is 0.3636, which is greater than the typical significance level of 0.05, we fail to reject the null hypothesis. This suggests that the residuals of the model are normally distributed. The Box-Ljung test gave a p-value of 0.5445, which is greater than 0.05. This indicates that there is no significant autocorrelation in the residuals of the model.

SARIMA(1,0,2)X(2,1,1)

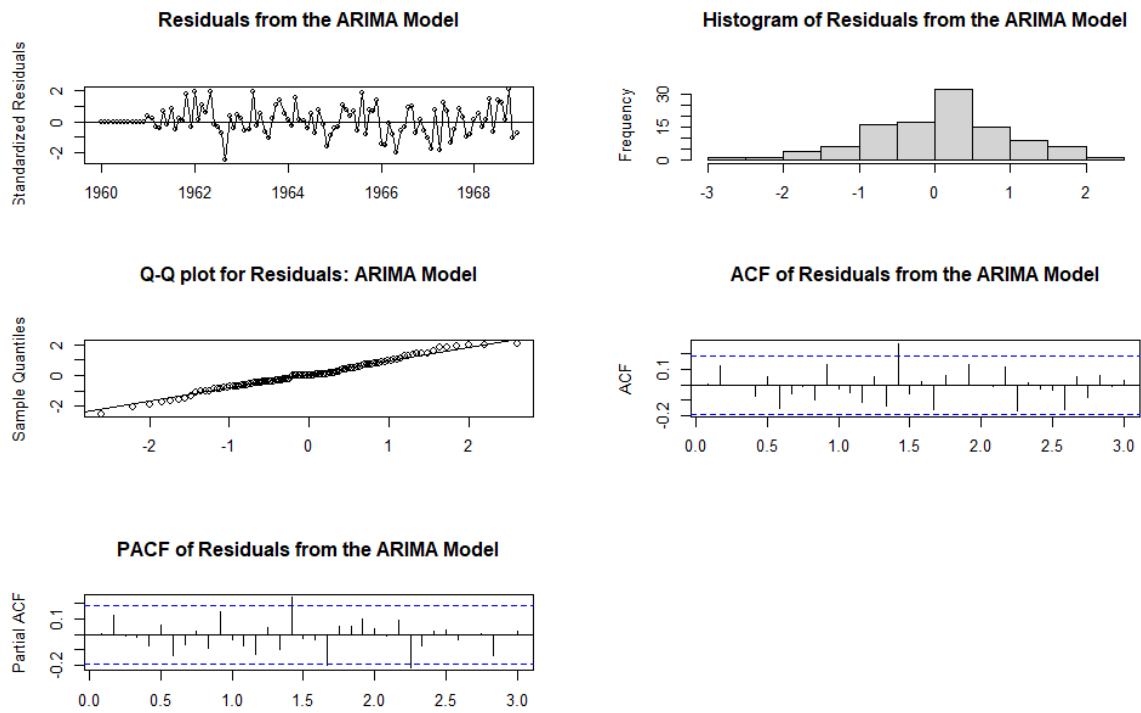


Figure 56: Residual Plots

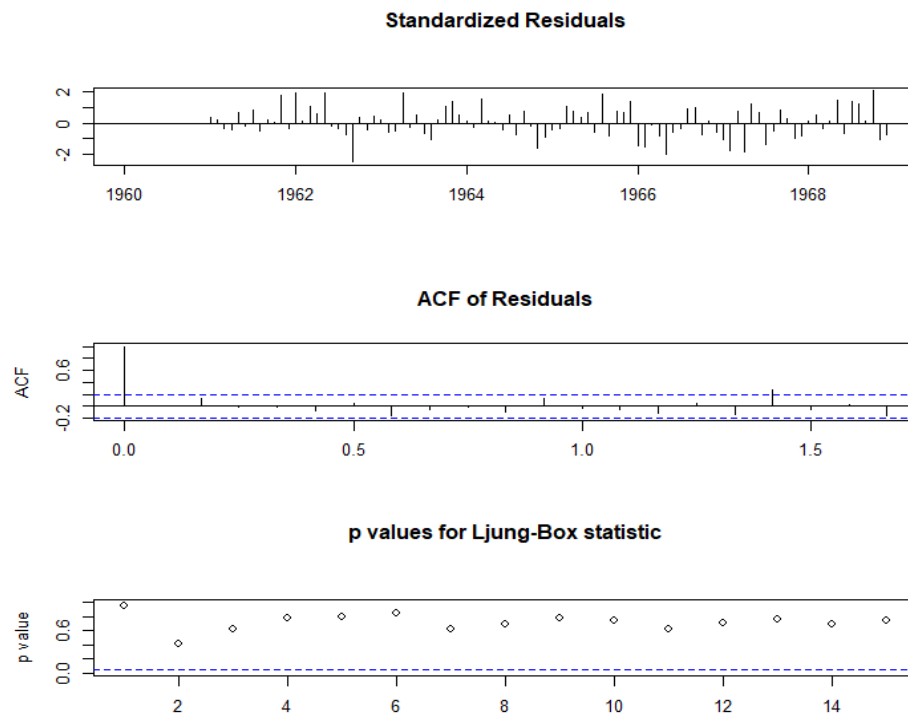


Figure 57: Residual Plots

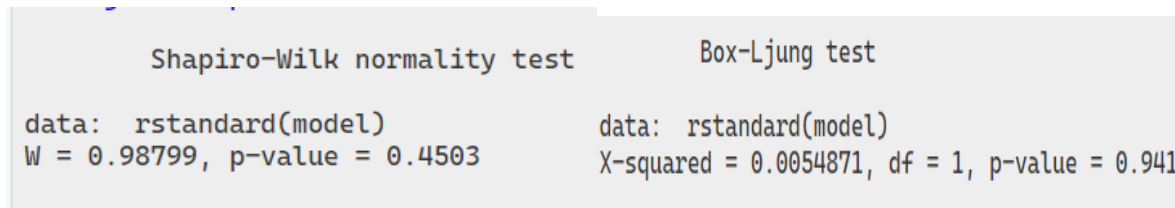


Figure 58: Shapiro-Wilk and Box-Ljung test results.

Inference: The residual plot of SARIMA(1,0,2)X(2,1,1) shows stationarity with a constant mean level, the histogram is most likely symmetric, Q-Q line fits all the data points which suggests normality. There is one significant late lag in both the ACF and PACF plots of residual. The Ljung-Box test has all the data points above the significance level showing no autocorrelation. Figure 58 shows the test result of the Shapiro-Wilk test and Ljung-Box test we get a p-value of 0.4503 from the Shapiro-Wilk test which is greater than the significance level of 0.05 suggesting normality. From the Ljung-Box test, we get a p-value of 0.941 which is again greater than the significance level and indicates that there is no significant autocorrelation in the residual of the model.

SARIMA(2,0,2)X(2,1,1)

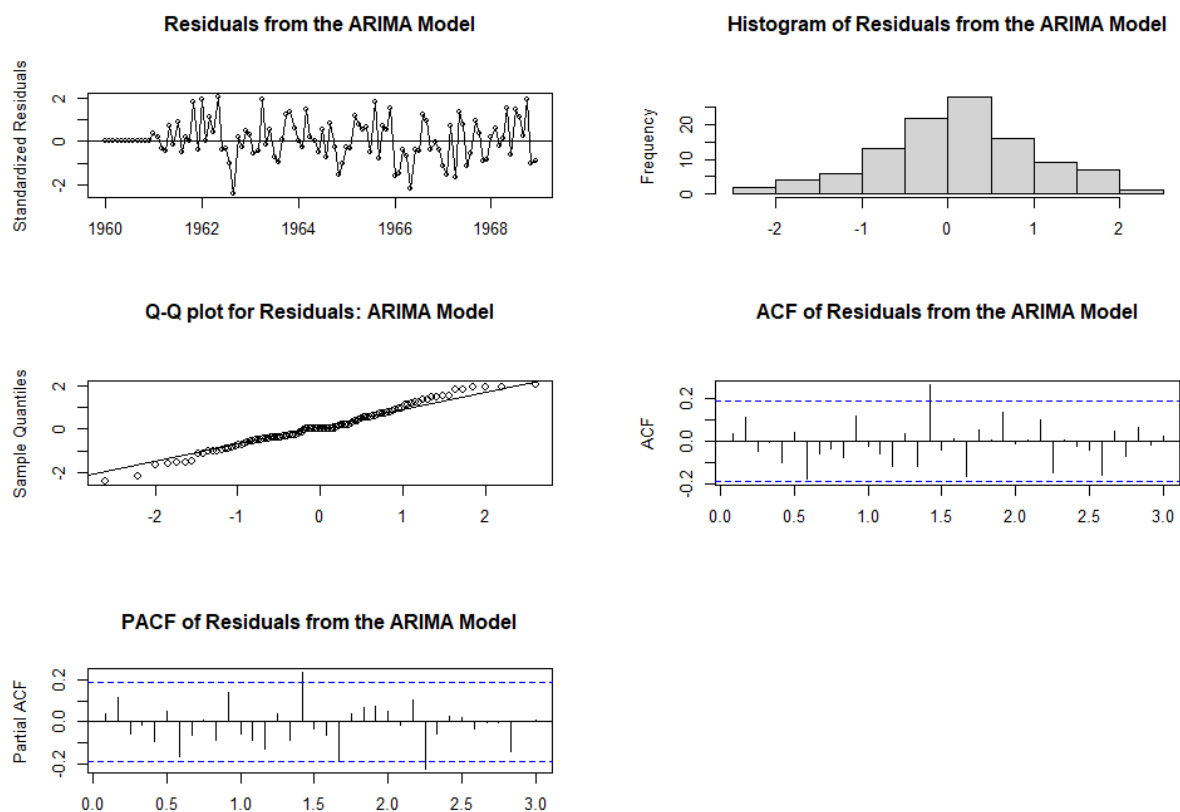


Figure 59: Residual Plots

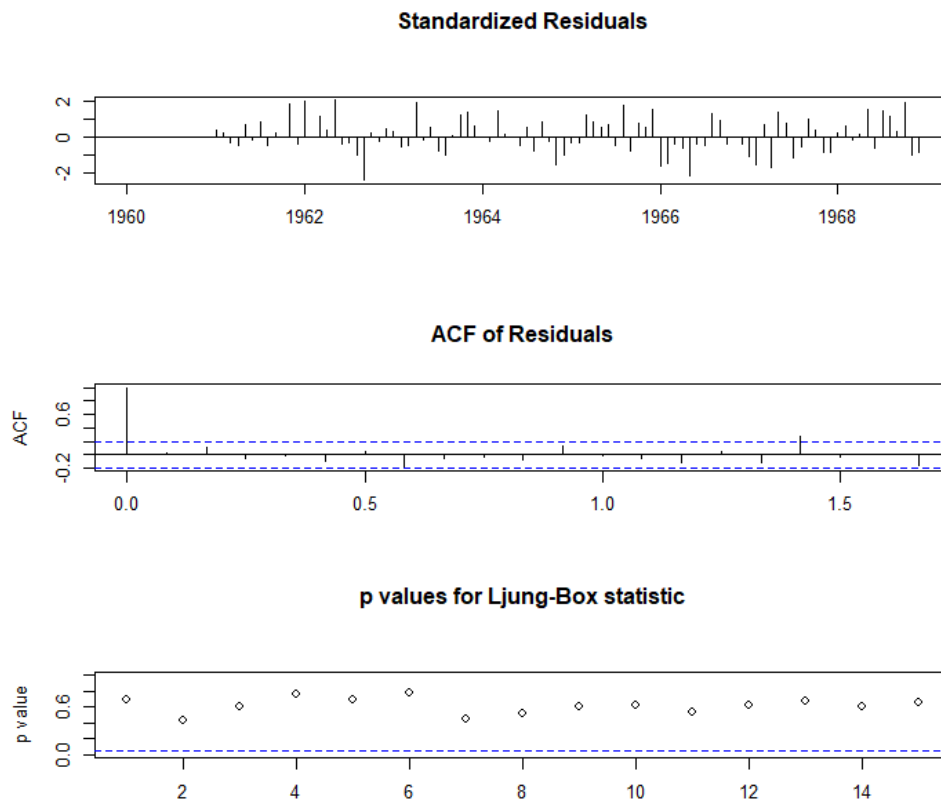


Figure 60: Residual Plots

Shapiro-Wilk normality test	Box-Ljung test
data: rstandard(model) W = 0.98548, p-value = 0.2919	data: rstandard(model) X-squared = 0.15162, df = 1, p-value = 0.697

Figure 61: Shapiro-Wilk and Box-Ljung test values

Inference: The residual plot of SARIMA(2,0,2)X(2,1,1) shows stationarity with a constant mean level, the histogram is most likely symmetric, the Q-Q line is not able to fit all the data points there is a tail on both ends. There is one significant late lag in both the ACF and PACF plots of residual. The Ljung-Box test has all the data points above the significance level showing no autocorrelation. Figure 61 shows the test result of the Shapiro-Wilk test and Ljung-Box test we get a p-value of 0.2919 from the Shapiro-Wilk test which is greater than the significance level of 0.05 suggesting normality. From the Ljung-Box test, we get a p-value of 0.697 which is again greater than the significance level and indicates that there is no significant autocorrelation in the residual of the model.

After residual analysis of those three models, we find that SARIMA(1,0,2)x(2,1,1) is the best model as it has the highest p-values for both the Shapiro-Wilk test (0.4503) and the Ljung-Box test (0.941), indicating the best adherence to normality and no autocorrelation in the residuals.

Based on model fitting, we got SARIMA(1,0,2)x(2,1,1) as the best model as well. Overall, this model provides the most reliable and statistically sound results based on the diagnostic tests, residual analysis, and coefficient test and error measures.

OVER PARAMETERIZING

SARIMA(0,0,2)X(2,1,1)

Parameter Estimation

Model	ML	CSS	CSS-ML
SARIMA(0,0,2)X(2,1,1)	ma1, ma2, sar1, sar2, sma1 significant.	ma1, sma1 significant.	ma1, ma2, sar1, sar2, sma1 significant.

Table 2

Residual analysis

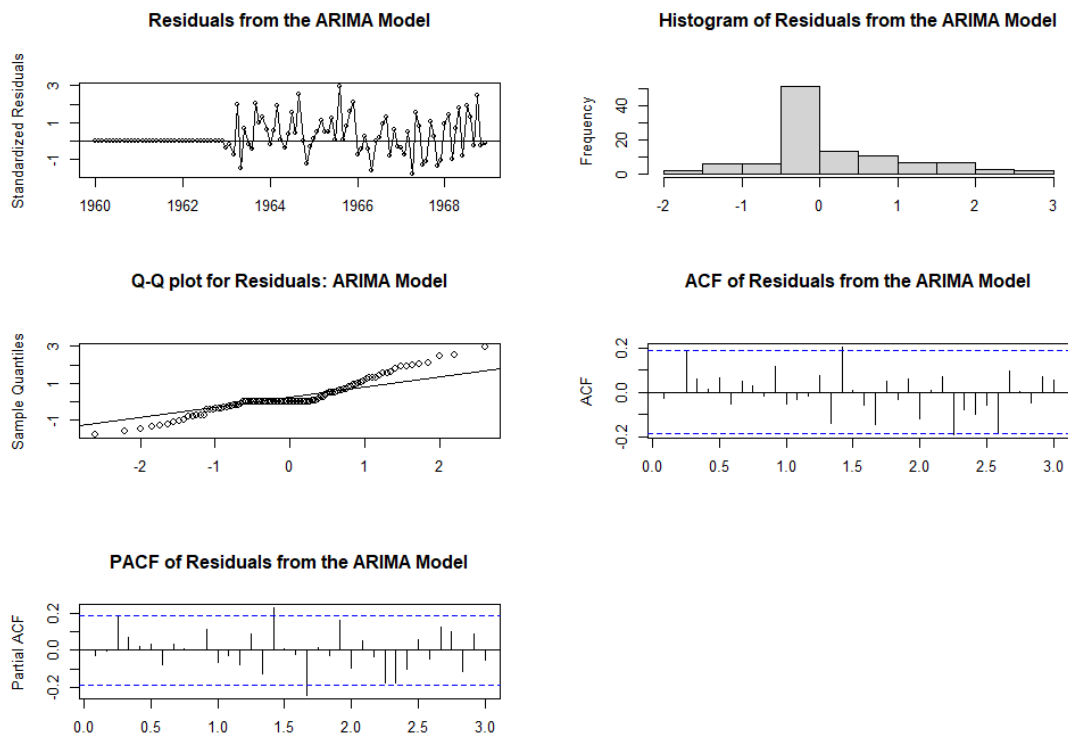


Figure 62: Residual Plots

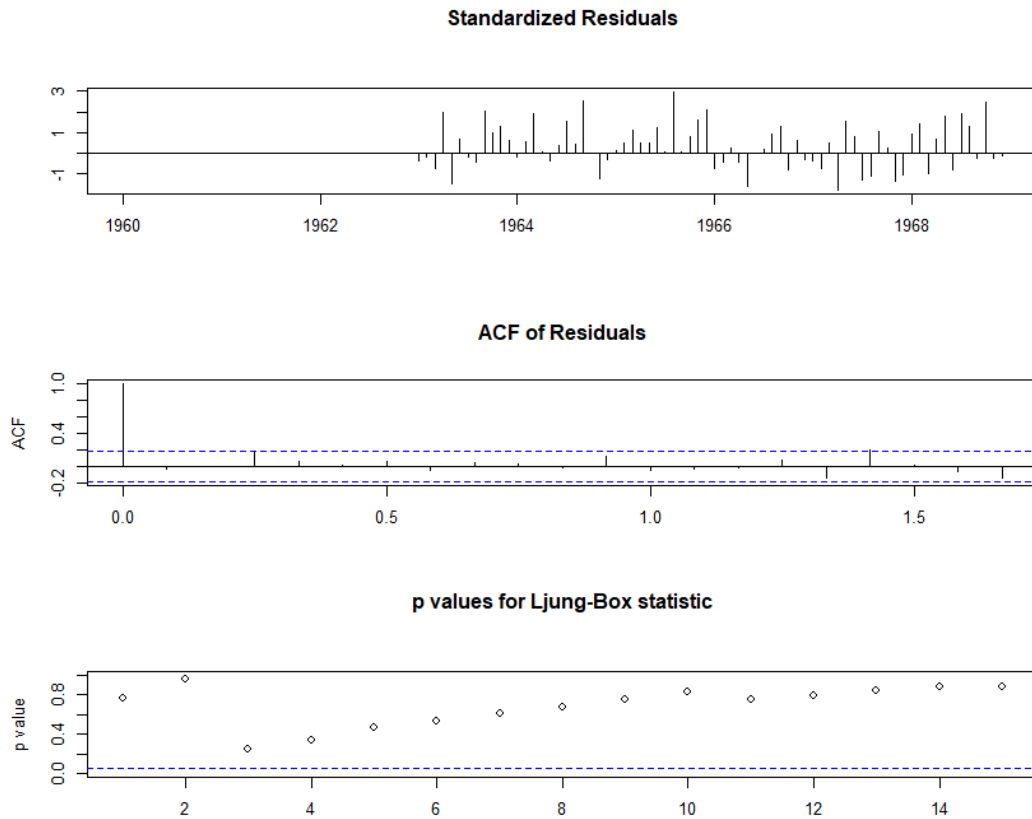


Figure 63: Residual Plots.

Shapiro-Wilk normality test	Box-Ljung test
data: rstandard(model) W = 0.92514, p-value = 1.328e-05	data: rstandard(model) X-squared = 0.089719, df = 1, p-value = 0.7645

Figure 64: Shapiro-Wilk and Box-Ljung test results.

Inference: Table 2 shows that all the coefficients are significant as per ML and CSS-ML methods. Figures 62 and 63, the time series plot of residuals seems stationary. The histogram is slightly right-skewed and does not show symmetry. The Q-Q line is not able to fit all the data points. ACF and PACF plots of the residuals have a near-miss lag and one significant late lag in PACF whereas two significant late lags in ACF plot. Figure 64 shows the test result of the Shapiro-Wilk and Ljung-Box test, the Shapiro-Wilk test gave a p-value of 1.328e-05, which is much less than the typical significance level of 0.05, this suggests that the residuals of the model are not normally distributed. The Box-Ljung test gave a p-value of 0.7645, which is greater than 0.05. This indicates that there is no significant autocorrelation in the residuals of the model.

SARIMA(1,0,3)X(2,1,1)

Parameter Estimation

Model	ML	CSS	CSS-ML
SARIMA(1,0,3)X(2,1,1)	ar1, ma1, sar1, sma1 significant.	ar1, ma1, sar1, sar2 significant.	ar1, ma1, sar1, sma1 significant.

Table 3

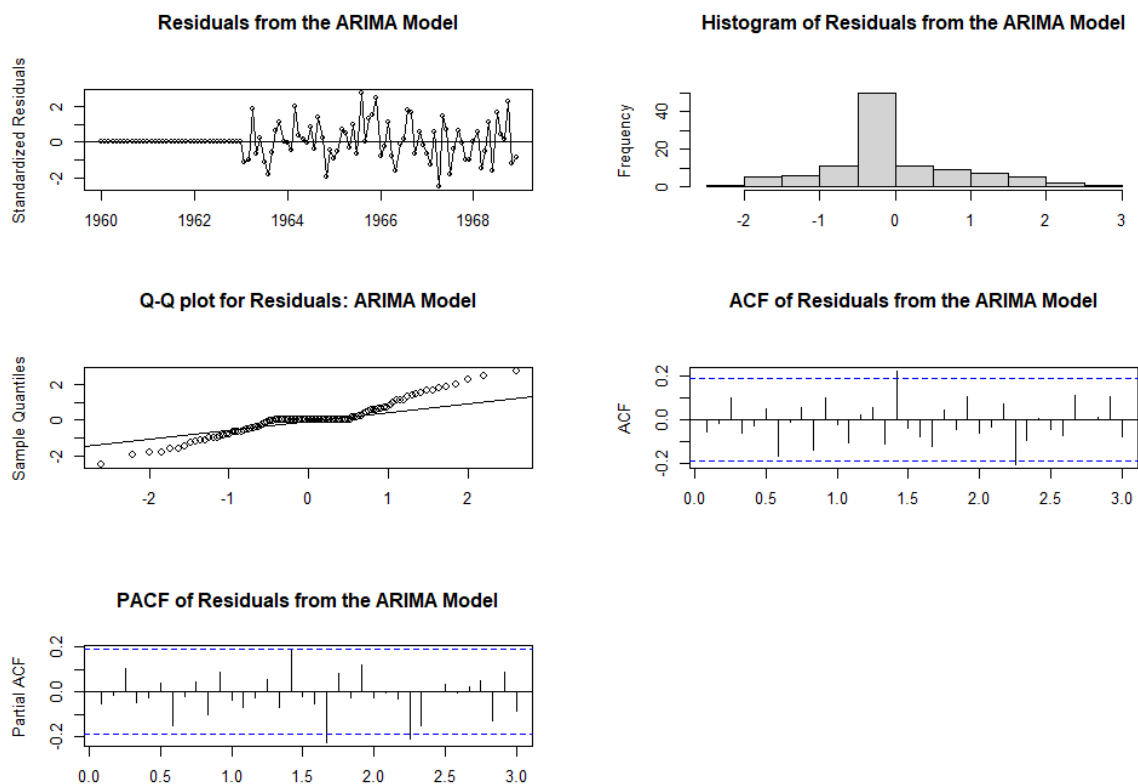


Figure 65: Residual Plots.

Shapiro-Wilk normality test	Box-Ljung test
<pre>data: rstandard(model) W = 0.93404, p-value = 4.442e-05</pre>	<pre>data: rstandard(model) X-squared = 0.29947, df = 1, p-value = 0.5842</pre>

Figure 66: Shapiro-Wilk and Ljung-Box test results.

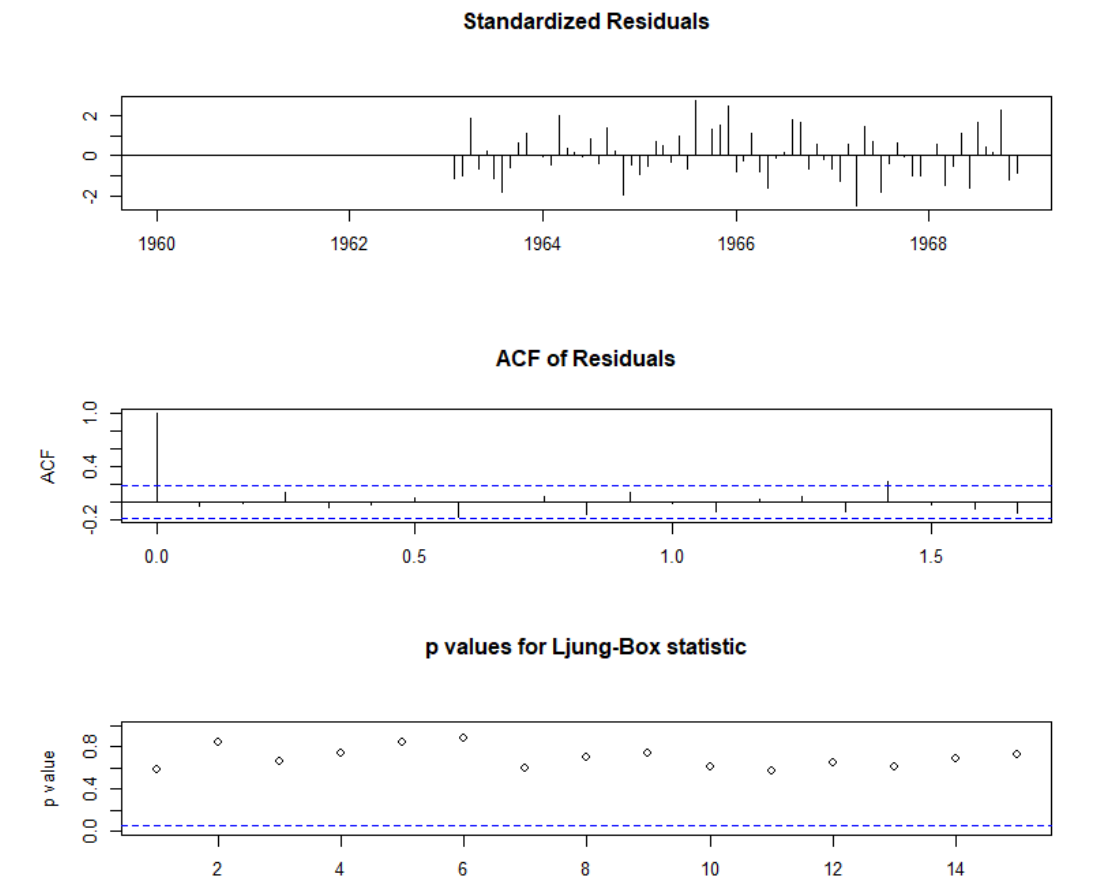


Figure 67: Residual Plots

Inference: Table 3 shows that ML and CSS-ML both have significant coefficients for ar1, ma1, sar1, and sma1. Whereas for CSS significant coefficients include ar1, ma1, sar1, and sar2. Figures 65 and 67 show the residual plots, time series plot of residuals seems to be stationary. The histogram seems symmetric, Q-Q line doesn't fit all the data points. In the ACF and PACF plots, we can see that the residuals of the model have one late significant lag. In the Ljung-Box plot, all the data points are above the significant level indicating no significant autocorrelation. From Figure 66 we got the test result of Shapiro-Wilk and Ljung-Box, Shapiro-Wilk test yielded a p-value of $4.442e-05$, which is much less than the typical significance level of 0.05. This suggests that the residuals of the model are not normally distributed. Whereas, the Box-Ljung test resulted in a p-value of 0.5842, which is greater than 0.05. This indicates that there is no significant autocorrelation present in the residuals of the model.

After over-parameterizing $\text{SARIMA}(1,0,2)\times(2,1,1)$, we find that the over-parameterized models aren't as well fitted as $\text{SARIMA}(1,0,2)\times(2,1,1)$. Therefore, for forecasting, we will use this model.

Result

FORECASTING

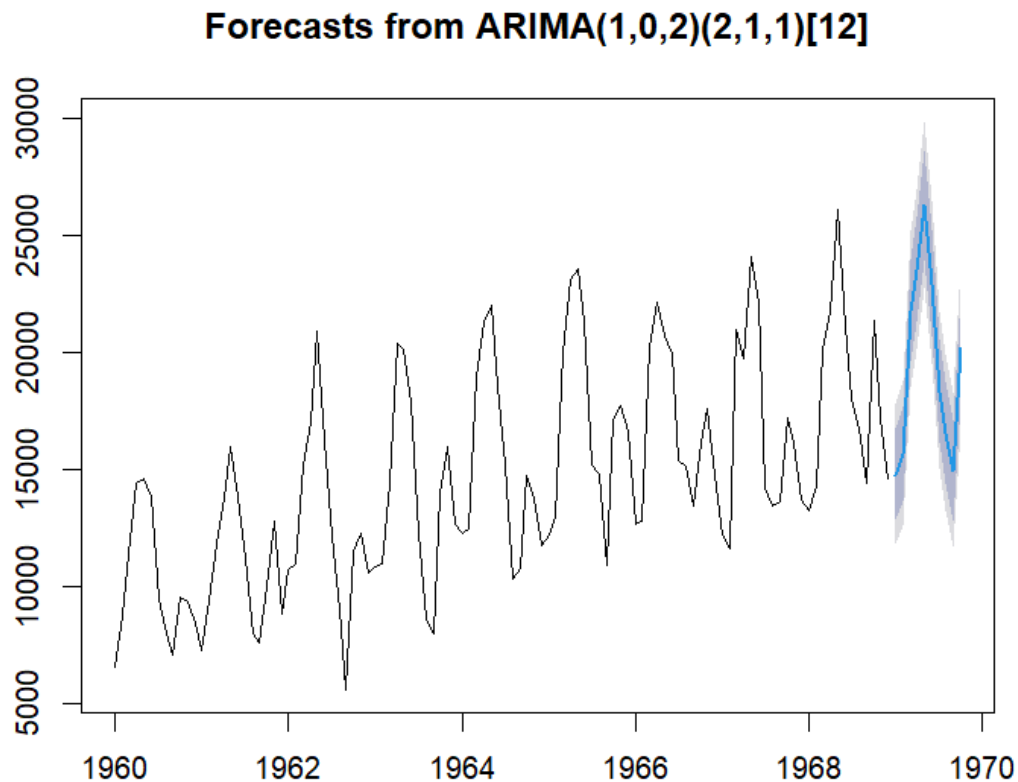


Figure 68: Forecasting for the 10 points ahead.

Point Forecast: These are the predicted values for each month starting from January 1969 to October 1969.

Prediction Intervals:

80% Confidence Intervals (Lo 80, Hi 80): These intervals are narrower and represent the range within which we expect 80% of future observations to fall.

95% Confidence Intervals (Lo 95, Hi 95): These intervals are wider and represent the range within which we expect 95% of future observations to fall.

Month	Point Forecast	Lo 80	Hi 80	Lo 95	Hi 95
Jan 1969	14734.53	12813.45	16707.15	11818.87	17770.96
Feb 1969	15621.60	13623.62	17672.15	12588.73	18777.61
Mar 1969	21750.16	19586.09	23958.22	18459.26	25144.04
Apr 1969	23476.49	21255.13	25740.75	20097.46	26955.95
May 1969	26261.29	23964.74	28598.77	22766.35	29852.01
Jun 1969	22924.38	20669.83	25224.22	19495.65	26459.14
Jul 1969	18325.97	16153.93	20550.83	15026.90	21748.77
Aug 1969	16520.51	14375.48	18722.85	13264.84	19910.45
Sept 1969	14888.13	12770.59	17067.82	11676.80	18245.14
Oct 1969	20182.86	17904.57	22513.86	16721.17	23767.99

Table 4: Forecast of Car sales for Jan 1969 to Oct 1969.

From January to May 1969, car sales are expected to have increased, with a peak predicted for May at 26,261.29 units. A possible seasonal pattern in car sales is indicated by the small fall in the anticipated values from June onwards.

Conclusion

Since this dataset is based on car sales, it is expected that the sales will go up as time progresses due to advancements in technology. The best model which is chosen for the forecast is SARIMA(1,0,2)x(2,1,1) since the original dataset has a seasonal component and this is confirmed by the original plot as well as the ACF plot. Following is the summary of reasons why the chosen model is the best-fitting model for the forecast:

1. Appears as the best model in EACF (top left model).
2. Appears as the best model in the BIC table.
3. Most of the coefficients are significant for this model.
4. Consistently appears as the second-best model in AIC & BIC scores.
5. Has the least error measures among all of the other possible models.
6. Finally, it also has strong support from the residuals as well.

Research states that the seasonal component in car sales is due to the end of the financial year and end-of-year deals. Overall, the obtained forecast is satisfactory however it is to be noted that this forecast is made under normal circumstances, if there is a strong unknown influencing factor in the upcoming future like the pandemic, then this forecast would not be accurate, therefore this report contains limitations.

Reference

Dataset used:

<https://data.world/perceptron/monthly-car-sales-quebec-1960>

Reading Material

Module 1:

https://rmit.instructure.com/courses/124176/files/36179093/download?download_frd=1

https://rmit.instructure.com/courses/124176/files/37065155/download?download_frd=1

Module 2:

https://rmit.instructure.com/courses/124176/files/37079754/download?download_frd=1

https://rmit.instructure.com/courses/124176/files/36178912/download?download_frd=1

Module 3:

https://rmit.instructure.com/courses/124176/files/36179092/download?download_frd=1

https://rmit.instructure.com/courses/124176/files/37162689/download?download_frd=1

Module 4:

https://rmit.instructure.com/courses/124176/files/36179098/download?download_frd=1

https://rmit.instructure.com/courses/124176/files/37260017/download?download_frd=1

Module 5:

https://rmit.instructure.com/courses/124176/files/36179219/download?download_frd=1

https://rmit.instructure.com/courses/124176/files/37445945/download?download_frd=1

Module 6:

https://rmit.instructure.com/courses/124176/files/36178867/download?download_frd=1

https://rmit.instructure.com/courses/124176/files/37562403/download?download_frd=1

Module 7:

https://rmit.instructure.com/courses/124176/files/36179239/download?download_frd=1

https://rmit.instructure.com/courses/124176/files/37887877/download?download_frd=1

Module 8:

https://rmit.instructure.com/courses/124176/files/36179169/download?download_frd=1

https://rmit.instructure.com/courses/124176/files/38018442/download?download_frd=1

R code reference:

https://rmit.instructure.com/courses/124176/files/37079799/download?download_frd=1

https://rmit.instructure.com/courses/124176/files/37172225/download?download_frd=1

https://rmit.instructure.com/courses/124176/files/37257888/download?download_frd=1

https://rmit.instructure.com/courses/124176/files/37445488/download?download_frd=1

https://rmit.instructure.com/courses/124176/files/37562032/download?download_frd=1

https://rmit.instructure.com/courses/124176/files/37748470/download?download_frd=1

https://rmit.instructure.com/courses/124176/files/37976026/download?download_frd=1

https://rmit.instructure.com/courses/124176/files/38104655/download?download_frd=1

Sort Score Function:

https://rmit.instructure.com/courses/124176/files/36179115/download?download_frd=1

Appendix

Link to the R codes used for the assignment

<https://drive.google.com/file/d/1PtaTcfnXg-XCdp8GXkIhEmanZ4hoMIz4/view?usp=sharing>

Necessary packages

```
rm(list=ls())
install.packages("TSA")
library(TSA)
install.packages("fUnitRoots")
library(fUnitRoots)
install.packages("FSAdata")
library(FSAdata)
library(tseries)
library(lmtest)
library(forecast)
library(dplyr)
```

Function for sorting AIC and BIC scores

```
# Function for sorting AIC and BIC output
sort.score <- function(x, score = c("bic", "aic")){
  if (score == "aic"){
    x[with(x, order(AIC)),]
  } else if (score == "bic") {
    x[with(x, order(BIC)),]
  } else {
    warning('score = "x" only accepts valid arguments ("aic","bic")')
  }
}
```

Function for stationarity test

```
# Stationarity test function
stationarity_tests ← function(data) {

  adf_result ← adf.test(data)
  cat("Augmented Dickey-Fuller (ADF) Test:\n")
  print(adf_result)

  kpss_result ← kpss.test(data)
  cat("\nKwiatkowski-Phillips-Schmidt-Shin (KPSS) Test:\n")
  print(kpss_result)

  pp_result ← pp.test(data)
  cat("\nPhillips-Perron (PP) Test:\n")
  print(pp_result)
}
```

Function for residual analysis

```
# Function for residual analysis
diagnostic_plots ← function(model) {
  # Set up the plotting area
  par(mfrow = c(3, 2))

  # Standardized Residuals Plot
  plot(rstandard(model),
       ylab = 'Standardized Residuals', type = 'o',
       main = "Residuals from the ARIMA Model")
  abline(h = 0)

  # Histogram of Standardized Residuals
  hist(rstandard(model), xlab = 'Standardized Residuals',
       main = "Histogram of Residuals from the ARIMA Model")

  # Q-Q Plot for Residuals
  qqnorm(rstandard(model), main = "Q-Q plot for Residuals: ARIMA Model")
  qqline(rstandard(model))

  # Shapiro-Wilk Normality Test
  shapiro_result ← shapiro.test(rstandard(model))
  print(shapiro_result)

  # ACF of Residuals
  acf(rstandard(model), lag.max = 36,
      main = "ACF of Residuals from the ARIMA Model")

  # PACF of Residuals
  pacf(rstandard(model), lag.max = 36,
      main = "PACF of Residuals from the ARIMA Model")

  # Ljung-Box Test
  ljung_box_result ← Box.test(rstandard(model), type = "Ljung-Box")
  print(ljung_box_result)

  # Time Series Diagnostic Plots
  tsdiag(model, gof = 15, omit.initial = FALSE)
}
```

Function for Coefficient test

```
# Function for coef test
fit_sarima_models <- function(ts_data, orders, seasonal_orders, period) {

  # Define a helper function to fit and test a model
  fit_and_test <- function(order, seasonal_order, method) {
    tryCatch({
      model <- Arima(ts_data, order=order, seasonal=list(order=seasonal_order, period=period), method=method)
      test_results <- coefest(model)
      return(test_results)
    }, error = function(e) {
      cat("Error fitting model with order =", paste(order, collapse=","),
          "and seasonal order =", paste(seasonal_order, collapse=","),
          "using method =", method, "\n")
      return(NULL)
    })
  }

  # Initialize a list to store results
  results <- list()

  # Loop through each combination of orders and methods
  for (order in orders) {
    for (seasonal_order in seasonal_orders) {
      for (method in c("ML", "CSS", "CSS-ML")) {
        model_name <- paste0("SARIMA(", paste(order, collapse=","), ")x(", paste(seasonal_order, collapse=","),
          cat("\nFitting model:", model_name, "\n")
          results[[model_name]] <- fit_and_test(order, seasonal_order, method)
        }
      }
    }
  }

  return(results)
}
```

Output of coefficient test

```
Model: SARIMA(2,0,4)x(2,1,1)_12.ML

z test of coefficients:

      Estimate Std. Error z value Pr(>|z|)
ar1   0.101409   0.063798  1.5895   0.11194
ar2   0.898548   0.063793 14.0854 < 2.2e-16 ***
ma1   0.186196   0.123118  1.5123   0.13045
ma2  -0.693517   0.116007 -5.9783 2.255e-09 ***
ma3  -0.080082   0.113404 -0.7062   0.48009
ma4  -0.199560   0.111180 -1.7949   0.07267 .
sar1   0.319669   0.124448  2.5687   0.01021 *
sar2   0.050904   0.139485  0.3649   0.71515
sma1 -0.973364   0.173983 -5.5946 2.211e-08 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Model: SARIMA(2,0,4)x(2,1,1)_12.CSS

z test of coefficients:

      Estimate Std. Error z value Pr(>|z|)
ar1   0.6862573   0.3210534  2.1375   0.03256 *
ar2   0.2985968   0.3169730  0.9420   0.34618
ma1  -0.3836516   0.3161729 -1.2134   0.22497
ma2  -0.3332898   0.2493324 -1.3367   0.18131
ma3  -0.0122584   0.1322237 -0.0927   0.92613
ma4  -0.1040429   0.1045292 -0.9953   0.31957
sar1  -0.5782380   0.3129840 -1.8475   0.06467 .
sar2  -0.2949597   0.1829404 -1.6123   0.10689
sma1 -0.0036813   0.3644232 -0.0101   0.99194
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Model: SARIMA(2,0,4)x(2,1,1)_12.CSS-ML

z test of coefficients:

	Estimate	Std. Error	z value	Pr(> z)
ar1	0.915213	0.416725	2.1962	0.02808 *
ar2	0.079994	0.417689	0.1915	0.84812
ma1	-0.655044	1.093165	-0.5992	0.54903
ma2	-0.061146	1.582501	-0.0386	0.96918
ma3	-0.087929	1.235702	-0.0712	0.94327
ma4	-0.075359	0.389126	-0.1937	0.84644
sar1	-0.445476	0.679719	-0.6554	0.51222
sar2	-0.239258	0.393812	-0.6075	0.54349
sma1	-0.053453	1.430490	-0.0374	0.97019

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Model: SARIMA(1,0,1)x(2,1,1)_12.ML

z test of coefficients:

	Estimate	Std. Error	z value	Pr(> z)
ar1	0.99993085	0.00093592	1068.3937	< 2.2e-16 ***
ma1	-0.80029201	0.09914381	-8.0720	6.914e-16 ***
sar1	0.30283764	0.12223055	2.4776	0.01323 *
sar2	0.09071423	0.13462476	0.6738	0.50042
sma1	-0.97774739	0.14890933	-6.5661	5.166e-11 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Model: SARIMA(1,0,1)x(2,1,1)_12.CSS

z test of coefficients:

	Estimate	Std. Error	z value	Pr(> z)
ar1	0.983351	0.016226	60.6041	< 2.2e-16 ***
ma1	-0.833603	0.105257	-7.9197	2.381e-15 ***
sar1	-0.413136	0.193514	-2.1349	0.03277 *
sar2	-0.180978	0.120785	-1.4983	0.13404
sma1	-0.175487	0.225051	-0.7798	0.43553

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Model: SARIMA(1,0,1)x(2,1,1)_12.CSS-ML

z test of coefficients:

	Estimate	Std. Error	z value	Pr(> z)
ar1	0.9998698	0.0014104	708.9484	< 2.2e-16 ***
ma1	-0.7997540	0.0993173	-8.0525	8.111e-16 ***
sar1	0.3022150	0.1235105	2.4469	0.01441 *
sar2	0.0906248	0.1349077	0.6718	0.50174
sma1	-0.9697994	0.1612590	-6.0139	1.811e-09 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Model: SARIMA(1,0,2)x(2,1,1)_12.ML

z test of coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
ar1	0.99996051	0.00051204	1952.8809	< 2.2e-16	***
ma1	-0.76182376	0.09609268	-7.9280	2.227e-15	***
ma2	-0.08638884	0.09937132	-0.8694	0.38465	
sar1	0.28658852	0.12304665	2.3291	0.01985	*
sar2	0.08987136	0.13574318	0.6621	0.50793	
sma1	-0.97562081	0.15585533	-6.2598	3.855e-10	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Model: SARIMA(1,0,2)x(2,1,1)_12.CSS

z test of coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
ar1	0.98705233	0.00020682	4772.4747	< 2.2e-16	***
ma1	-0.78942469	0.11134005	-7.0902	1.339e-12	***
ma2	-0.31212698	0.13013029	-2.3986	0.0164591	*
sar1	-0.69783322	0.03290859	-21.2052	< 2.2e-16	***
sar2	-0.36250275	0.10068255	-3.6005	0.0003177	***
sma1	0.13741150	0.12671551	1.0844	0.2781833	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Model: SARIMA(1,0,2)x(2,1,1)_12.CSS-ML

z test of coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
ar1	0.99996051	0.00051204	1952.8809	< 2.2e-16	***
ma1	-0.76182376	0.09609268	-7.9280	2.227e-15	***
ma2	-0.08638884	0.09937132	-0.8694	0.38465	
sar1	0.28658852	0.12304665	2.3291	0.01985	*
sar2	0.08987136	0.13574318	0.6621	0.50793	
sma1	-0.97562081	0.15585533	-6.2598	3.855e-10	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Model: SARIMA(2,0,2)x(2,1,1)_12.ML

z test of coefficients:

	Estimate	Std. Error	z value	Pr(> z)
ar1	0.076884	0.051641	1.4888	0.13654
ar2	0.922727	0.051549	17.8999	< 2.2e-16 ***
ma1	0.183347	0.104157	1.7603	0.07836 .
ma2	-0.816638	0.101129	-8.0752	6.736e-16 ***
sar1	0.295231	0.129687	2.2765	0.02282 *
sar2	0.101785	0.137190	0.7419	0.45813
sma1	-0.959047	0.200309	-4.7878	1.686e-06 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Model: SARIMA(2,0,2)x(2,1,1)_12.CSS

z test of coefficients:

	Estimate	Std. Error	z value	Pr(> z)
ar1	0.68193	0.32419	2.1035	0.03542 *
ar2	0.29961	0.31800	0.9422	0.34609
ma1	-0.38699	0.29370	-1.3176	0.18763
ma2	-0.38990	0.24807	-1.5717	0.11601
sar1	-0.56200	0.31140	-1.8048	0.07111 .
sar2	-0.26996	0.18337	-1.4722	0.14097
sma1	-0.02354	0.35331	-0.0666	0.94688

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Model: SARIMA(1,0,0)x(2,1,1)_12.ML

z test of coefficients:

	Estimate	Std. Error	z value	Pr(> z)
ar1	0.612821	0.094109	6.5118	7.425e-11 ***
sar1	-1.052075	0.245711	-4.2818	1.854e-05 ***
sar2	-0.445874	0.115457	-3.8618	0.0001125 ***
sma1	0.683354	0.300199	2.2763	0.0228261 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Model: SARIMA(1,0,0)x(2,1,1)_12.CSS

z test of coefficients:

	Estimate	Std. Error	z value	Pr(> z)
ar1	0.683595	0.086052	7.9440	1.958e-15 ***
sar1	-0.691438	0.185072	-3.7360	0.0001869 ***
sar2	-0.373093	0.137472	-2.7140	0.0066484 **
sma1	0.228842	0.195006	1.1735	0.2405900

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Model: SARIMA(1,0,0)x(2,1,1)_12.CSS-ML

z test of coefficients:

	Estimate	Std. Error	z value	Pr(> z)
ar1	0.612542	0.094127	6.5076	7.637e-11 ***
sar1	-1.053415	0.245712	-4.2872	1.809e-05 ***
sar2	-0.446020	0.115383	-3.8655	0.0001108 ***
sma1	0.685015	0.300514	2.2795	0.0226388 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1


```
Model: SARIMA(2,0,0)x(2,1,1)_12.ML
```

```
z test of coefficients:
```

	Estimate	Std. Error	z value	Pr(> z)	
ar1	0.415681	0.099615	4.1729	3.008e-05	***
ar2	0.433247	0.101070	4.2866	1.814e-05	***
sar1	0.352496	0.284208	1.2403	0.21487	
sar2	0.021300	0.173358	0.1229	0.90221	
sma1	-0.832489	0.374912	-2.2205	0.02639	*

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
Model: SARIMA(2,0,0)x(2,1,1)_12.CSS
```

```
z test of coefficients:
```

	Estimate	Std. Error	z value	Pr(> z)	
ar1	0.45317	0.10780	4.2038	2.625e-05	***
ar2	0.32511	0.10603	3.0662	0.002168	**
sar1	-0.34984	0.27366	-1.2784	0.201108	
sar2	-0.21818	0.14942	-1.4601	0.144253	
sma1	-0.15123	0.30390	-0.4976	0.618740	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
Model: SARIMA(2,0,0)x(2,1,1)_12.CSS-ML
```

```
z test of coefficients:
```

	Estimate	Std. Error	z value	Pr(> z)	
ar1	0.394703	0.104005	3.7950	0.0001476	***
ar2	0.337681	0.099114	3.4070	0.0006568	***
sar1	-0.862583	0.436901	-1.9743	0.0483452	*
sar2	-0.350070	0.142548	-2.4558	0.0140569	*
sma1	0.507178	0.477167	1.0629	0.2878302	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Assigning variable names for aic, bic, error measure and diagnostic check

```
> # SARIMA(2,0,4)x(2,1,1)_12
> m3_204.salesML = Arima(Car_SalesTS_BC, order=c(2,0,4), seasonal=list(order=c(2,1,1), period=12), method="ML")
> m3_204.salesCSS = Arima(Car_SalesTS_BC, order=c(2,0,4), seasonal=list(order=c(2,1,1), period=12), method="CSS")
> # SARIMA(1,0,1)x(2,1,1)_12
> m3_101.salesML = Arima(Car_SalesTS_BC, order=c(1,0,1), seasonal=list(order=c(2,1,1), period=12), method="ML")
> m3_101.salesCSS = Arima(Car_SalesTS_BC, order=c(1,0,1), seasonal=list(order=c(2,1,1), period=12), method="CSS")
> # SARIMA(1,0,2)x(2,1,1)_12
> m3_102.salesML = Arima(Car_SalesTS_BC, order=c(1,0,2), seasonal=list(order=c(2,1,1), period=12), method="ML")
> m3_102.salesCSS = Arima(Car_SalesTS_BC, order=c(1,0,2), seasonal=list(order=c(2,1,1), period=12), method="CSS")
```

```

> # SARIMA(2,0,2)x(2,1,1)_12
> m3_202.salesML = Arima(Car_SalesTS_BC, order=c(2,0,2), seasonal=list(order=c(2,1,1), period=12), method="ML")
> m3_202.salesCSS = Arima(Car_SalesTS_BC, order=c(2,0,2), seasonal=list(order=c(2,1,1), period=12), method="CSS")
> # SARIMA(1,0,0)x(2,1,1)_12
> m3_100.salesML = Arima(Car_SalesTS_BC, order=c(1,0,0), seasonal=list(order=c(2,1,1), period=12), method="ML")
> m3_100.salesCSS = Arima(Car_SalesTS_BC, order=c(1,0,0), seasonal=list(order=c(2,1,1), period=12), method="CSS")
> # SARIMA(2,0,0)x(2,1,1)_12
> m3_200.salesML = Arima(Car_SalesTS_BC, order=c(2,0,0), seasonal=list(order=c(2,1,1), period=12), method="ML")
> m3_200.salesCSS = Arima(Car_SalesTS_BC, order=c(2,0,0), seasonal=list(order=c(2,1,1), period=12), method="CSS")

```

Residual Analysis

```

# Residual Analysis

# SARIMA(2,0,4)x(2,1,1)_12
diagnostic_plots(m3_204.salesML)

# SARIMA(1,0,1)x(2,1,1)_12
diagnostic_plots(m3_101.salesML)

# SARIMA(1,0,2)x(2,1,1)_12
diagnostic_plots(m3_102.salesML)

# SARIMA(2,0,2)x(2,1,1)_12
diagnostic_plots(m3_202.salesML)

# SARIMA(1,0,0)x(2,1,1)_12
diagnostic_plots(m3_100.salesML)
|
# SARIMA(2,0,0)x(2,1,1)_12
diagnostic_plots(m3_200.salesML)

```

SARIMA(2,0,4)x(2,1,1)_12

```

> diagnostic_plots(m3_204.salesML)

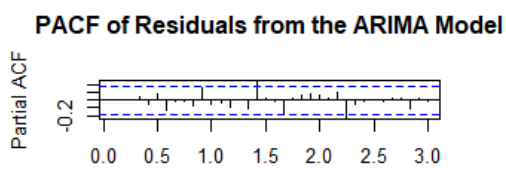
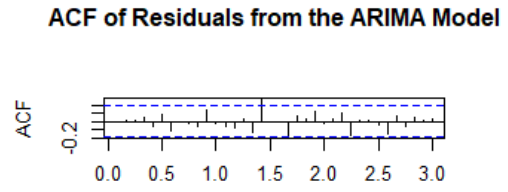
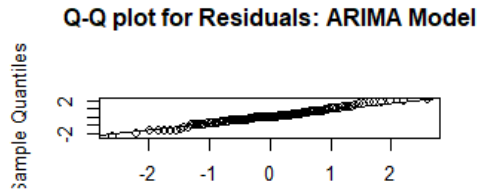
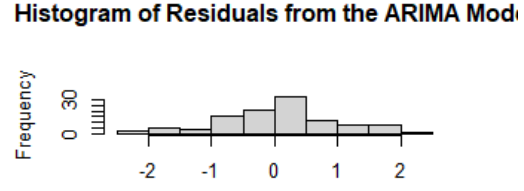
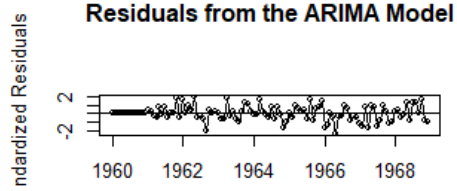
      Shapiro-Wilk normality test

data:  rstandard(model)
W = 0.98595, p-value = 0.3177

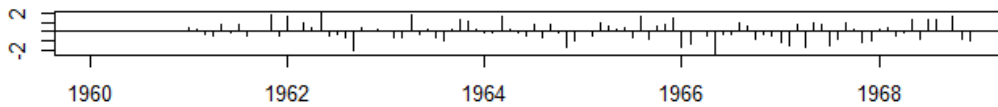
      Box-Ljung test

data:  rstandard(model)
X-squared = 1.2961e-08, df = 1, p-value = 0.9999

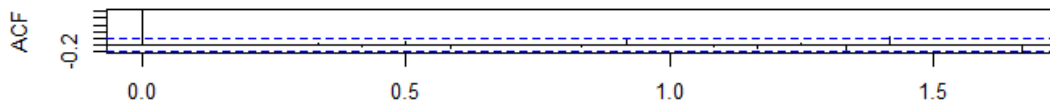
```



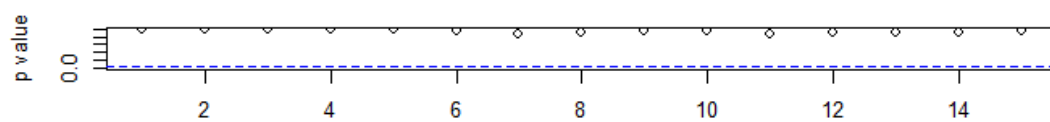
Standardized Residuals



ACF of Residuals



p values for Ljung-Box statistic



SARIMA(1,0,0)x(2,1,1)₁₂

```
> diagnostic_plots(m3_100.salesML)
```

Shapiro-Wilk normality test

data: rstandard(model)

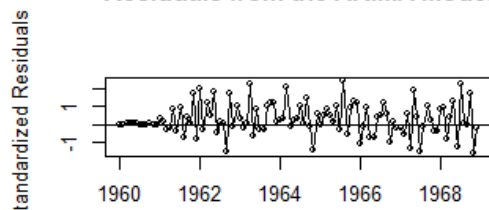
W = 0.97613, p-value = 0.04899

Box-Ljung test

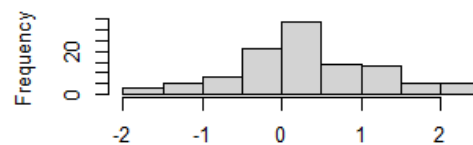
data: rstandard(model)

X-squared = 11.065, df = 1, p-value = 0.0008797

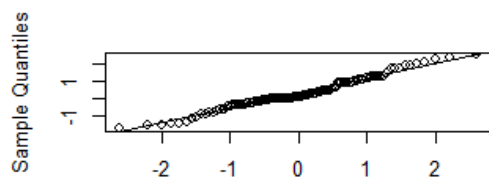
Residuals from the ARIMA Model



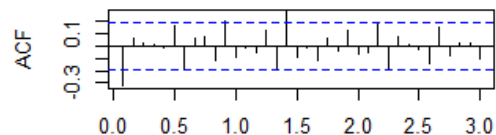
Histogram of Residuals from the ARIMA Model



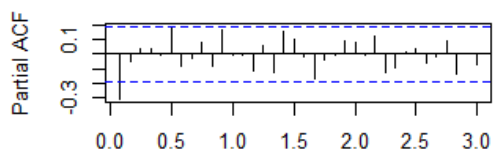
Q-Q plot for Residuals: ARIMA Model

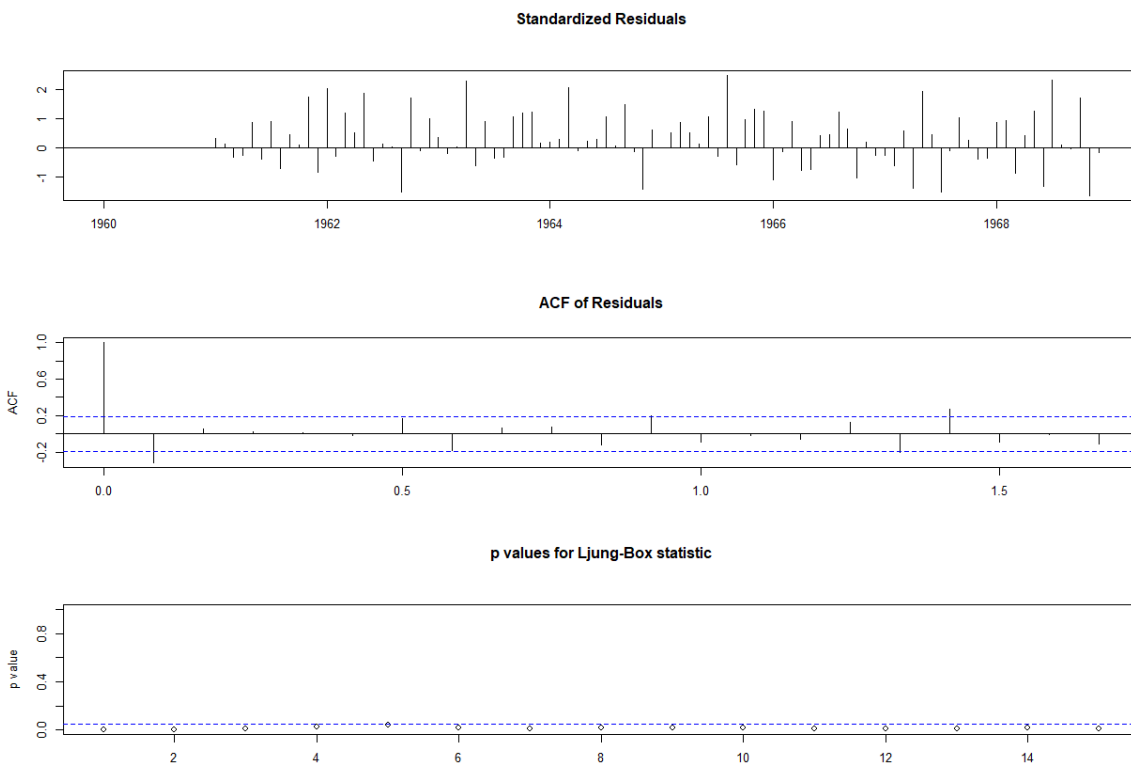


ACF of Residuals from the ARIMA Model



PACF of Residuals from the ARIMA Model





SARIMA(2,0,0)x(2,1,1)₁₂

```
> diagnostic_plots(m3_200.salesML)
```

Shapiro-Wilk normality test

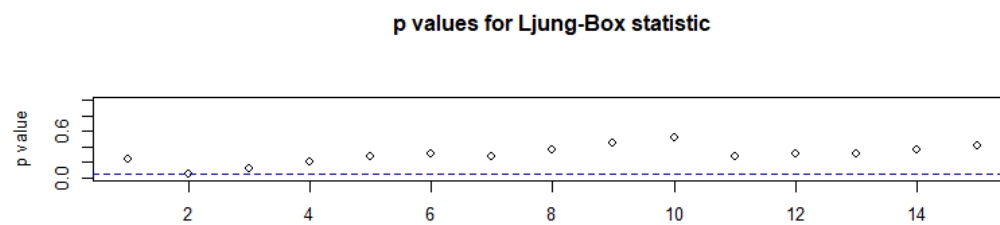
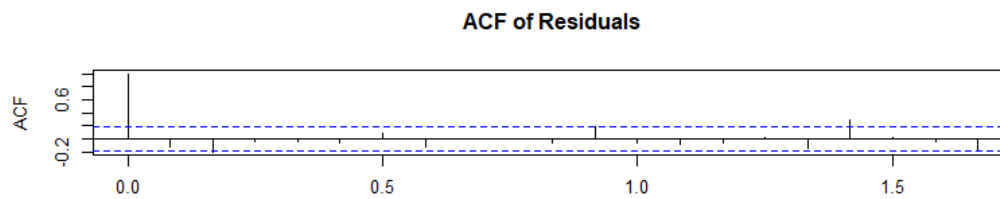
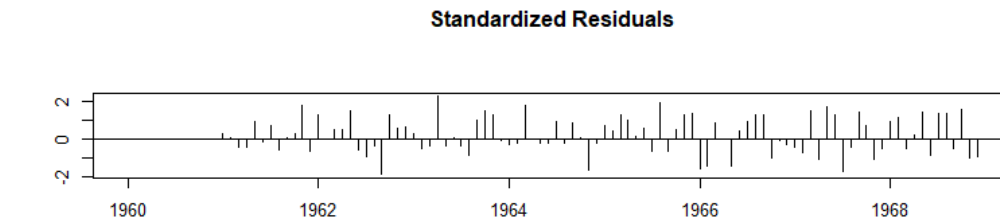
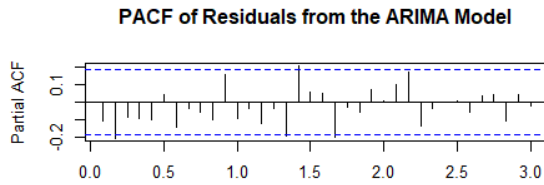
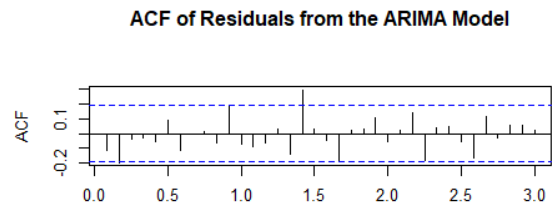
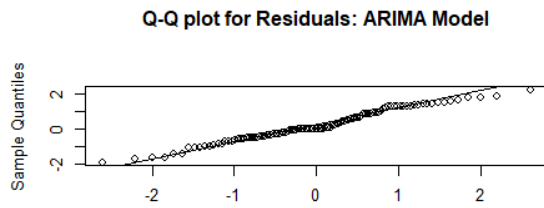
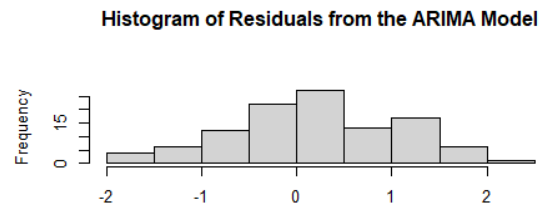
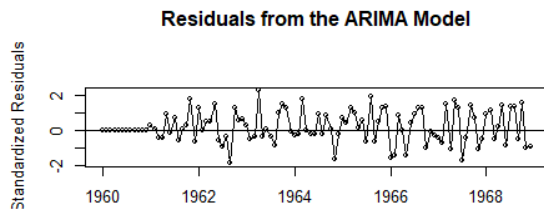
```
data: rstandard(model)
```

```
W = 0.97903, p-value = 0.08586
```

Box-Ljung test

```
data: rstandard(model)
```

```
X-squared = 1.4297, df = 1, p-value = 0.2318
```



Over-parameterized models

```
# After all the analysis SARIMA(1,0,2)x(2,1,1)_12 is the best model
# We will do over parameterization

orders <- list(c(0,0,2), c(1,0,3))
seasonal_orders <- list(c(2,1,1))
results <- fit_sarima_models(Car_SalesTS_BC, orders, seasonal_orders, period=12)

# Print results
for (name in names(results)) {
  cat("\nModel:", name, "\n")
  if (!is.null(results[[name]])) {
    print(results[[name]])
  } else {
    cat("No results for this model due to an error.\n")
  }
}

m3_002.salesCSS = Arima(Car_SalesTS_BC, order=c(0,0,2), seasonal=list(order=c(2,1,1), period=12),
  method="CSS")
m3_103.salesCSS = Arima(Car_SalesTS_BC, order=c(1,0,3), seasonal=list(order=c(2,1,1), period=12),
  method="CSS")
# SARIMA(0,0,2)x(2,1,1)
diagnostic_plots(m3_002.salesCSS)
# SARIMA(1,0,3)x(2,1,1)
diagnostic_plots(m3_103.salesCSS)
```

Model: SARIMA(0,0,2)x(2,1,1)_12.ML

z test of coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
ma1	0.255354	0.109211	2.3382	0.019378	*
ma2	0.261610	0.093637	2.7939	0.005208	**
sar1	0.735230	0.119467	6.1542	7.544e-10	***
sar2	0.261303	0.113581	2.3006	0.021414	*
sma1	-0.972322	0.157435	-6.1760	6.573e-10	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1					

Model: SARIMA(0,0,2)x(2,1,1)_12.CSS

z test of coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
ma1	0.22840	0.11465	1.9921	0.046356	*
ma2	0.18112	0.10402	1.7411	0.081665	.
sar1	0.16460	0.15291	1.0764	0.281743	
sar2	0.22130	0.11551	1.9158	0.055390	.
sma1	-0.47473	0.16812	-2.8238	0.004745	**

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1					

Model: SARIMA(0,0,2)x(2,1,1)_12.CSS-ML

z test of coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
ma1	0.25537	0.10922	2.3383	0.019374	*
ma2	0.26162	0.09364	2.7939	0.005207	**
sar1	0.73471	0.12038	6.1033	1.039e-09	***
sar2	0.26133	0.11358	2.3009	0.021400	*
sma1	-0.97040	0.15948	-6.0849	1.166e-09	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Model: SARIMA(1,0,3)x(2,1,1)_12.ML

z test of coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
ar1	0.99996668	0.00035569	2811.3170	< 2.2e-16	***
ma1	-0.76766591	0.10421930	-7.3659	1.760e-13	***
ma2	0.04355466	0.13726055	0.3173	0.75101	
ma3	-0.16241594	0.10639071	-1.5266	0.12686	
sar1	0.31601433	0.12336690	2.5616	0.01042	*
sar2	0.04348337	0.13895234	0.3129	0.75433	
sma1	-0.96848171	0.16469887	-5.8803	4.095e-09	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Model: SARIMA(1,0,3)x(2,1,1)_12.CSS

z test of coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
ar1	0.99269121	0.00015458	6421.8455	< 2.2e-16	***
ma1	-0.80893245	0.12081509	-6.6956	2.148e-11	***
ma2	-0.24194212	0.15419389	-1.5691	0.116630	
ma3	-0.06609851	0.12803061	-0.5163	0.605665	
sar1	-0.59011222	0.01982528	-29.7656	< 2.2e-16	***
sar2	-0.34751092	0.09841773	-3.5310	0.000414	***
sma1	0.04180984	0.12341535	0.3388	0.734780	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Model: SARIMA(1,0,3)x(2,1,1)_12.CSS-ML

z test of coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
ar1	0.99996668	0.00035569	2811.3170	< 2.2e-16	***
ma1	-0.76766591	0.10421930	-7.3659	1.760e-13	***
ma2	0.04355466	0.13726055	0.3173	0.75101	
ma3	-0.16241594	0.10639071	-1.5266	0.12686	
sar1	0.31601433	0.12336690	2.5616	0.01042	*
sar2	0.04348337	0.13895234	0.3129	0.75433	
sma1	-0.96848171	0.16469887	-5.8803	4.095e-09	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Forecast

```
# Forecast for SARIMA(1,0,2)x(2,1,1)_12
m3.CarSales = Arima(Car_SalesTS, order=c(1,0,2),
                    seasonal=list(order=c(2,1,1), period=12),
                    lambda = 0.8)
future = forecast(m3.CarSales, lambda = 0.8, h = 10)
future
plot(future)
```